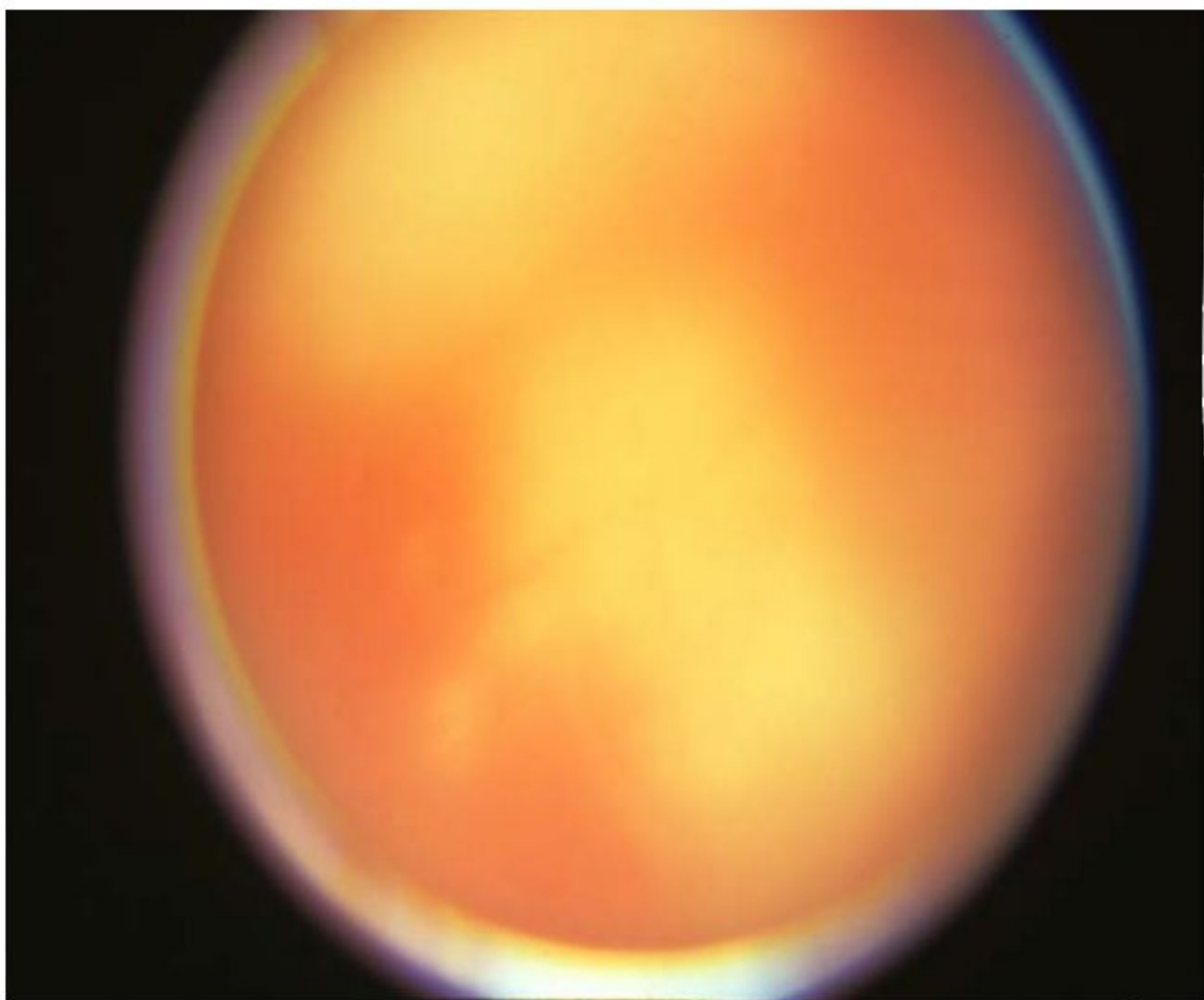


choroiditis patch { yellowish creamy patch }
+ vitritis { vitreous hazy } } => Head light in Fog



Mx ∴

- Prednisolone [1mg/kg] and tapered according to clinical response
- Pyrimethamin [follic acid antagonist] + Oral folic acid
 - ↳ retards thrombocytopenia leukopenia and folate deff.
- Sulfa diazine

Congenital TOXOPLASMOSES:



- Punched out
- Chorioretinal scar
- pigmented border → RPE proliferation
- @ MACULA → B/L

In Reactivation: Choroidal plaque adjacent scar

↓
Mx: Intravitreal therapy w/ Clindamycin
+ Dexamethasone

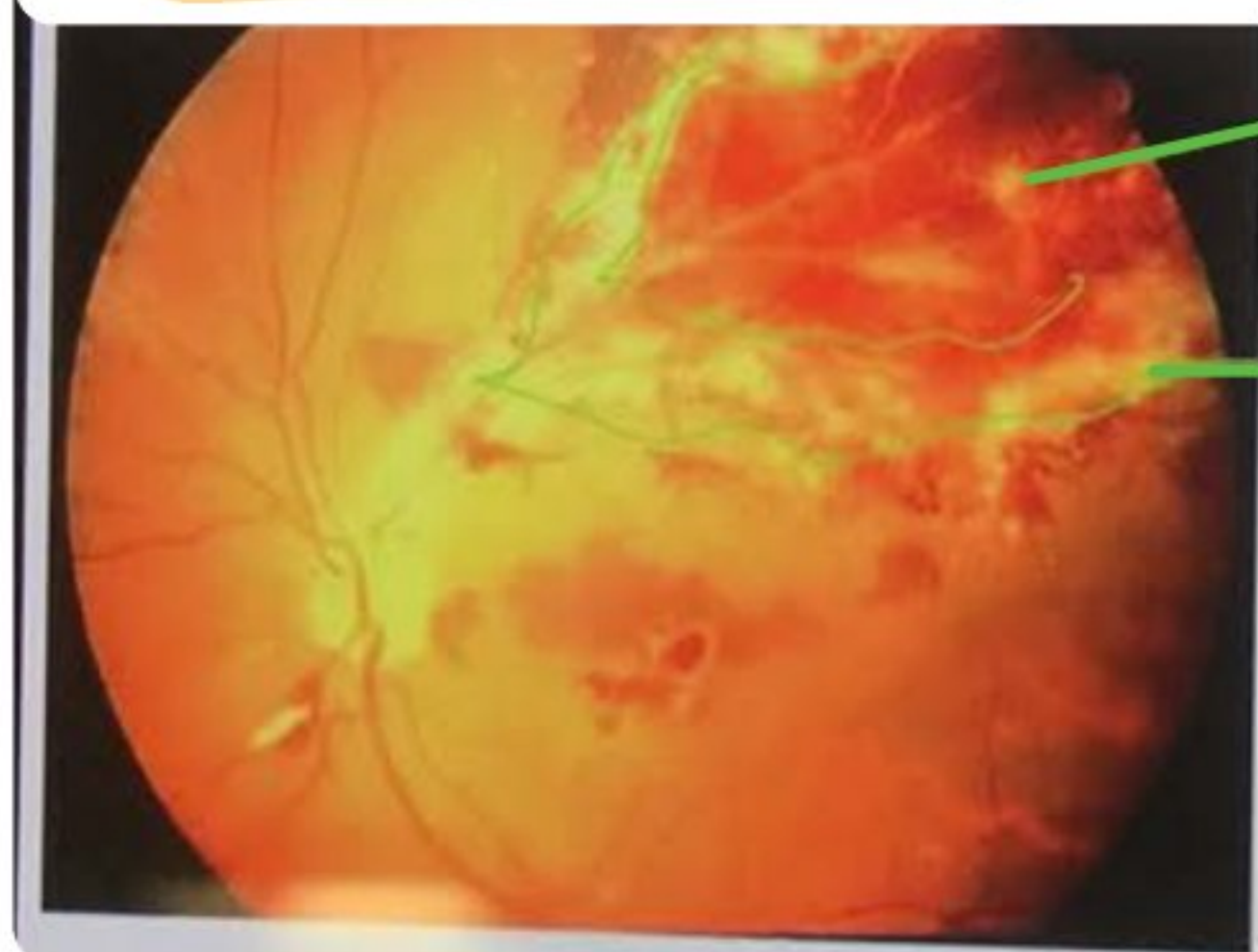
Investigation.

↳ TORCH serology IgM & IgG Ab

CMV Chorioretinitis

- MCC of posterior uveitis in immunocompromised patients (HIV $\Rightarrow$ in CD4 $< 50/\mu\text{L}$)
 - C/P : i) Retinal H'morages $\rightarrow$ Red
 &
 Retinal necrosis $\rightarrow$ white
- q: $\oplus$ Vitreous H'morages
q: $\oplus$ Perivascularitis
- 
Notespaedia.com
More than just notes

CMV CHORIORETINITIS



retinal H'mage

retinal necrosis

BUSH FIRE appearance / Pizza Pie appearance/ Sauce and cheese appearance

(MCC) A/c retinal necrosis in adults } Herpes Zoster Ophthalmicus

(MCC) A/c retinal necrosis in children } CMV HSV

Investigation:

Q: Most accurate & rapid method $\Rightarrow$ viral nucleic acid detection in intra ocular fluid.

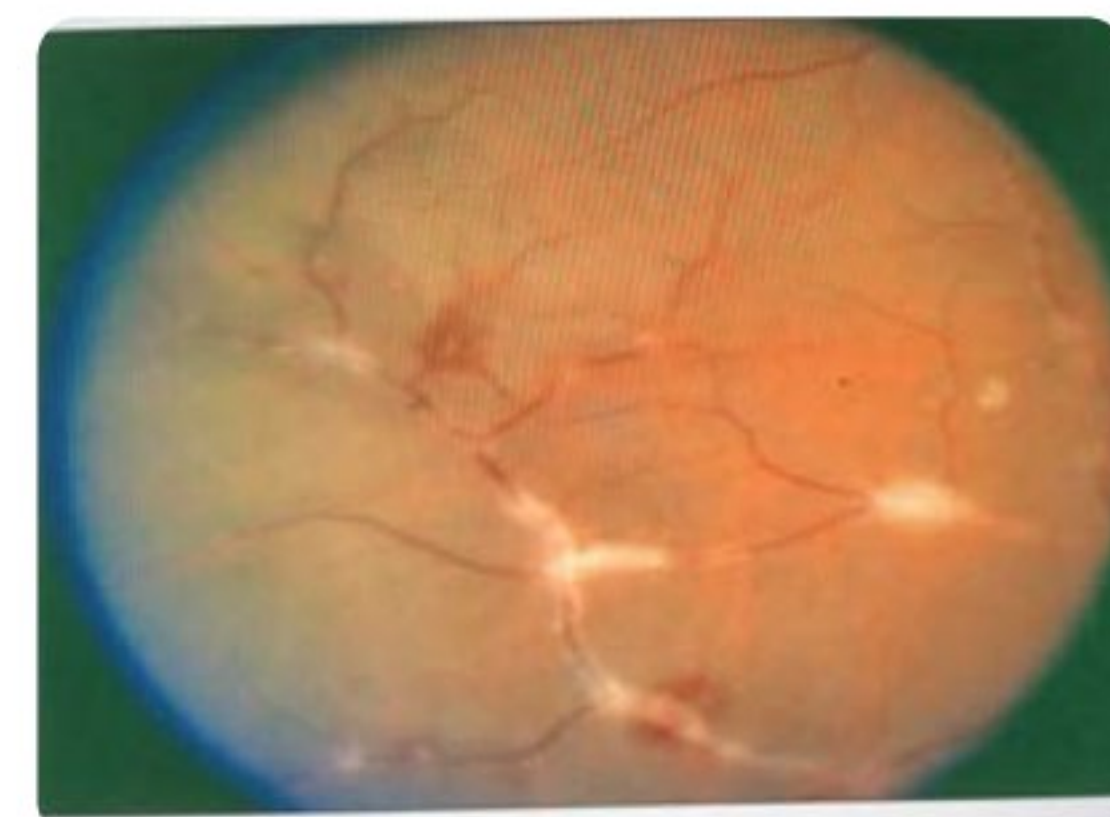
TOC: GANCYCLOVIR | for HIV associated CMV retinitis: HAART until CD4+ > 100-150 cells/mL

U/L $\Rightarrow$ intravitreal gancyclovir Implant {to avoid systemic toxicity}

B/L $\Rightarrow$ System IV Gancyclovir

SARCOIDOSIS: post. uveitis

- Granulomatous anterior uveitis is the most common
- Pre-Retinal nodule $\rightarrow$ LANDER'S SIGN
Inflammation of retinal vessel



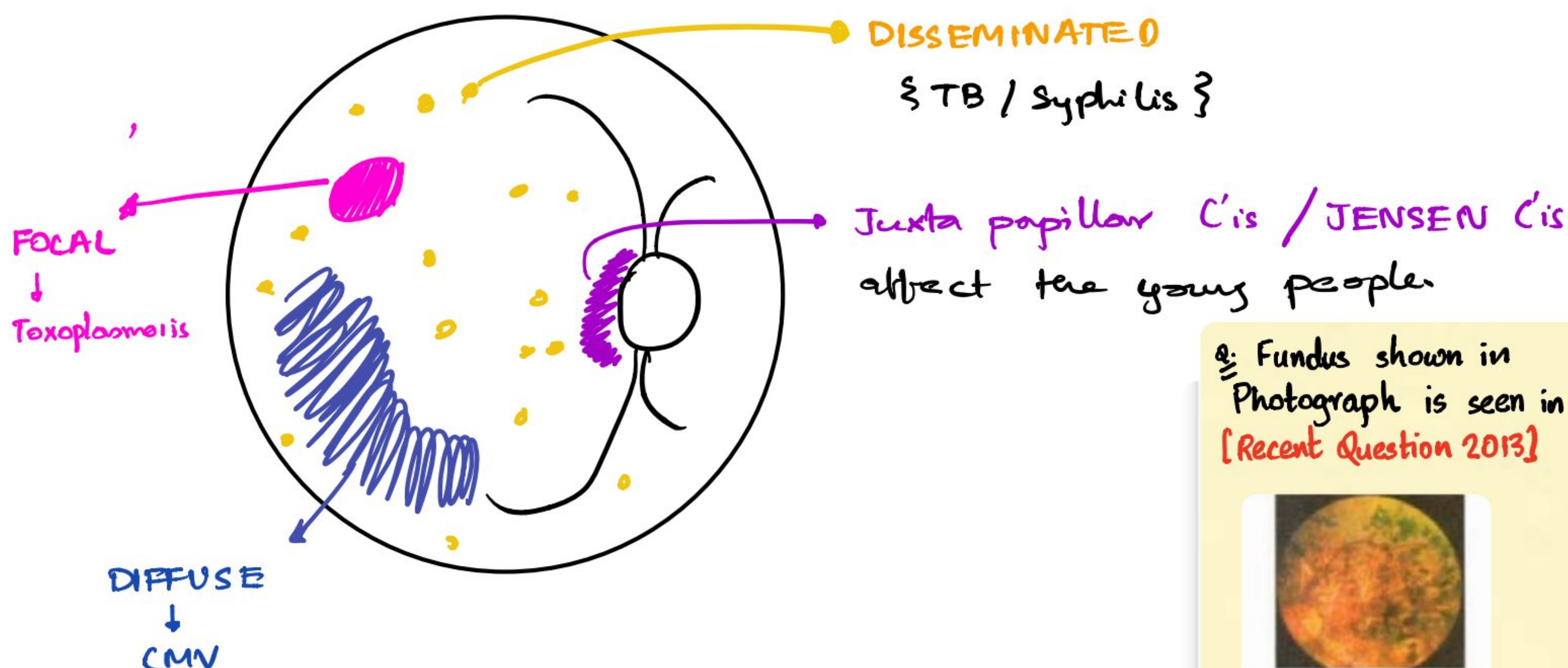
Candle wax dripping



Retinal vessel

Inflammatory exudates } CANDLE WAX DRIPPING

Types of Chorioiditis:



Q: Fundus shown in Photograph is seen in [Recent Question 2013]



- Cong. Histoplasmosis
- Cong. Syphilis
- Cong. Toxoplasmosis
- Cong. HIV

SYMPATHETIC OPHTHALMIA:

- B/L granulomatous Panuveitis
- affect the uninvolved & sympathising / non injured eye following penetrating injury / intraocular Sx to the exciting eye (injured eye)

Eye is immunoprivileged organ → SEQUESTERED Ag



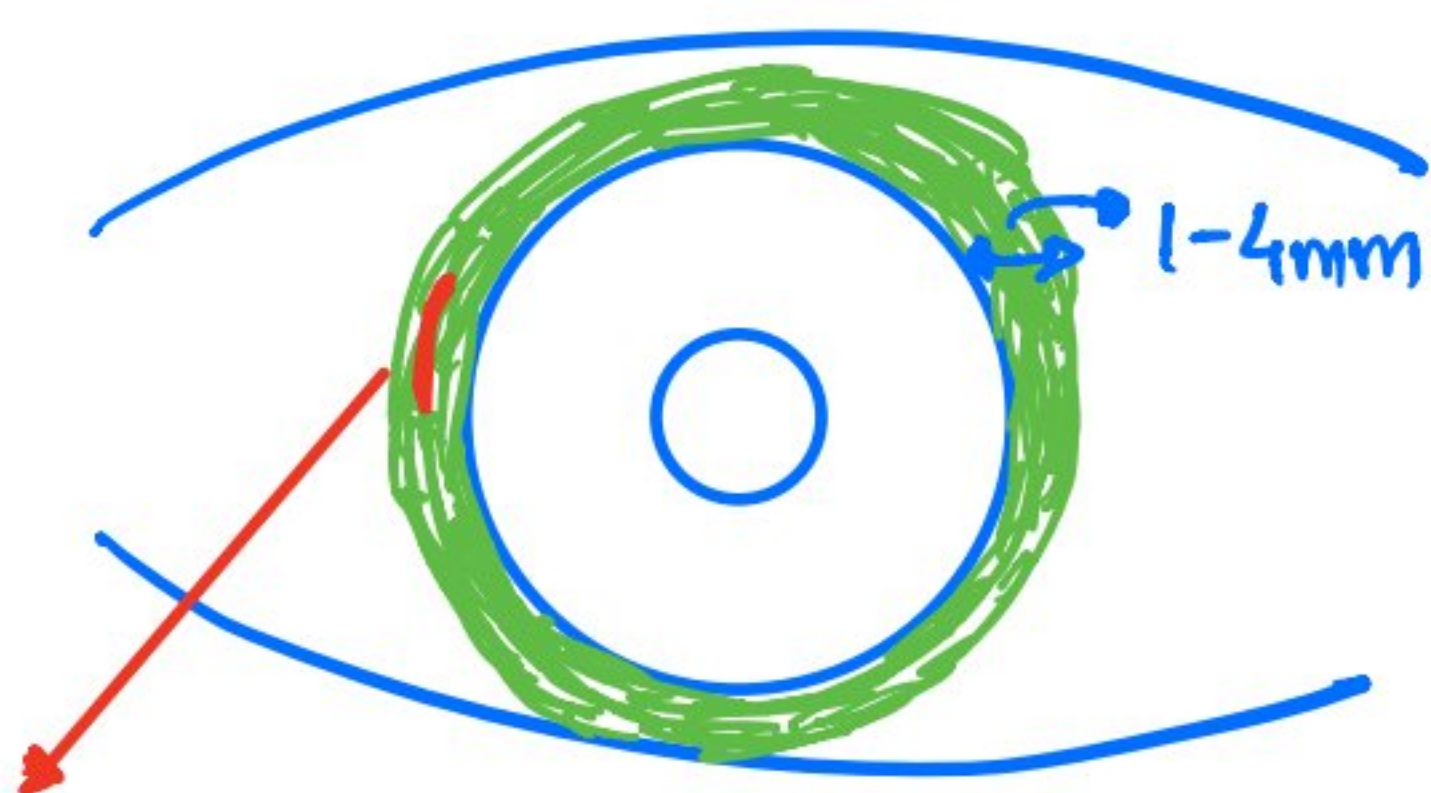
Retinal S. Antigen

→ during trauma gets exposed and body starts recognition this is a new Ag.

site of penetrating injury
depends on
i) Ciliary Region

AIMS
Danger ZONE
of the eye

>4mm for the Limbus

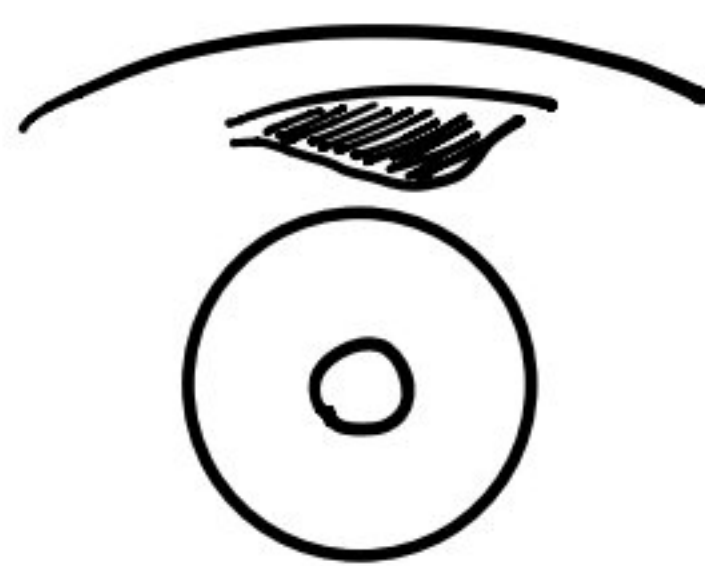


∴ all incision in IO surgery should be <1-4mm from Limbus.

Time taken by immune system to mount a response } 2wk - 3months after accident

Other risk factors:

- i) Limbal injury
- ii) Iris prolapse into the wound.



1st sign of S.O.

Retrolental flare in sympathising eye.



c. Muscle affected → blurred near vision.

in sympathizing eye

1st symptom.

Q: DALEN FUCH'S NODULE are s/i

- a) Corneal ulcer
- b) Retinitis pigmentosa
- c) Sympathetic ophthalmitis
- d) After cataract.

DALEN FUCH'S nodules → are s/i - VKH
• sympathetic ophthalmitis

Granulomatous Pan Uveitis

- Mutton fat KPs
- Iris nodule
- Dalen - fuch's nodules in choroid.

Mx :: Prevention of S.O

- Systemic steroid: Suppress the immune system
- Enucleation of exciting eye within 2wk of injury } old method

Rx: i) Systemic steroid { high dose ; long duration }

or
ii) Systemic immunosuppression :: by Azathioprine Mycophenolate.

VKH Syndrome :: B/L granulomatous panuveitis

(Vogt-Koyanagi Harada)

- Autoimmune not a/w trauma
- HLA DR4 association
- Japanese population

System involved ::

- 1) CNS → Meningismus (Neck stiffness without meningitis)
- 2) Ear → Vertigo ⊕, Tinnitus ⊕, Deafness ⊕
- 3) Skin/Hair → Vitiligo, Alopecia
Poliosis (Premature graying of lashes)

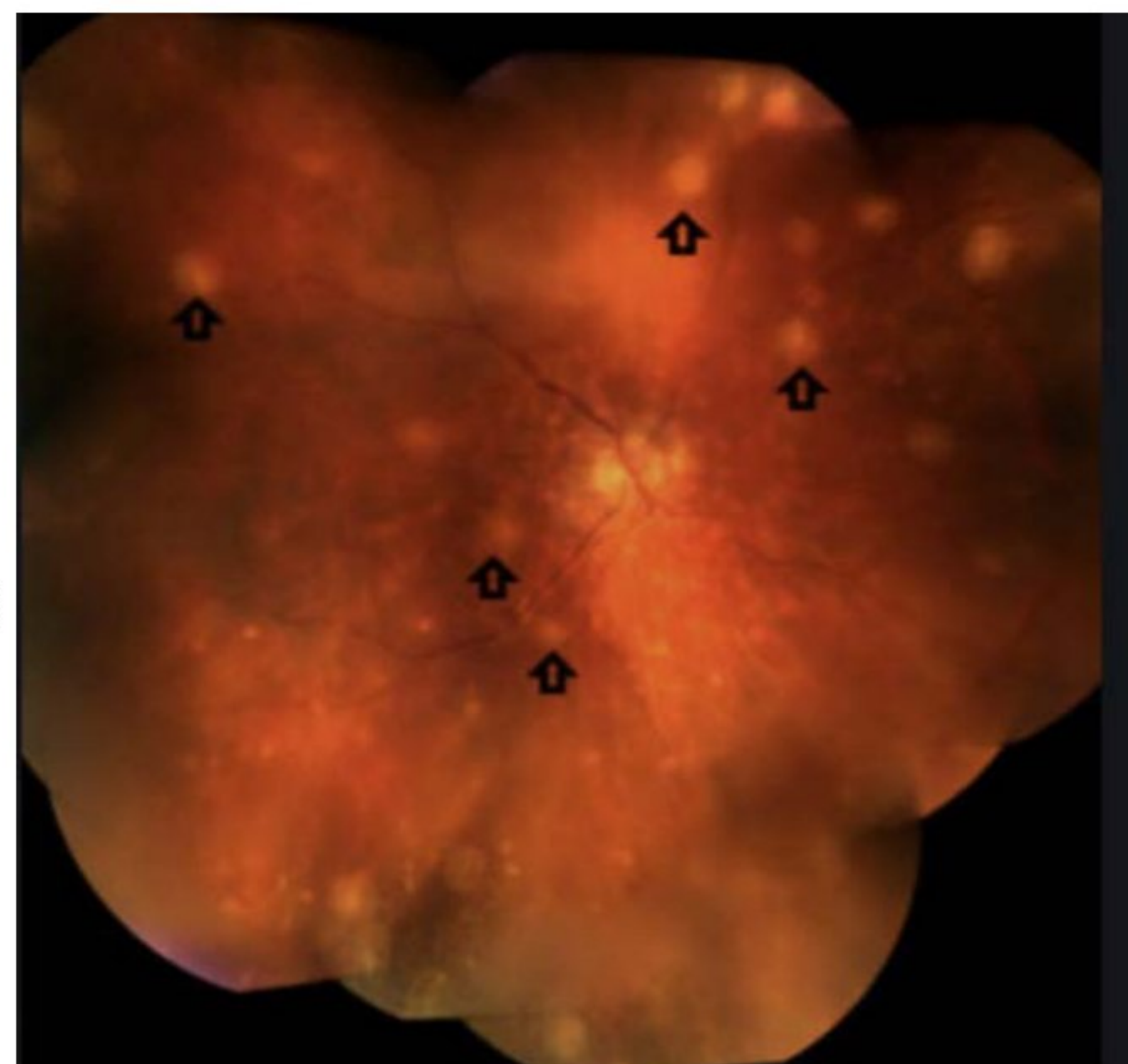
4) Eye feature

i) Perilimbal vitiligo ⇒ **SUGIURA SIGN**

ii) B/L granulomatous Pan. uveitis →
• Mutton fat KPs
• Iris nodules
• Dalen Fuch's nodules

5) Exclusive finding in eye ::

- i) Multiple serous retinal detachment



DALEN FUCHS NODULE

ii) Sunset glow app. of fundus.

Mx: Systemic steroids (high dose) to keep immune system is suppressed

Modified Diagnostic Criteria

for VKH Sx ::
~~~~~

1) Absence of H/o trauma

2) " of ocular d/s

3) B/L granulomatous panuveitis

4) Neurological & auditory manif.

5) Other finding: Alopecia

• Poliosis

• Vitiligo

### CHOROIDAL MELANOMA:

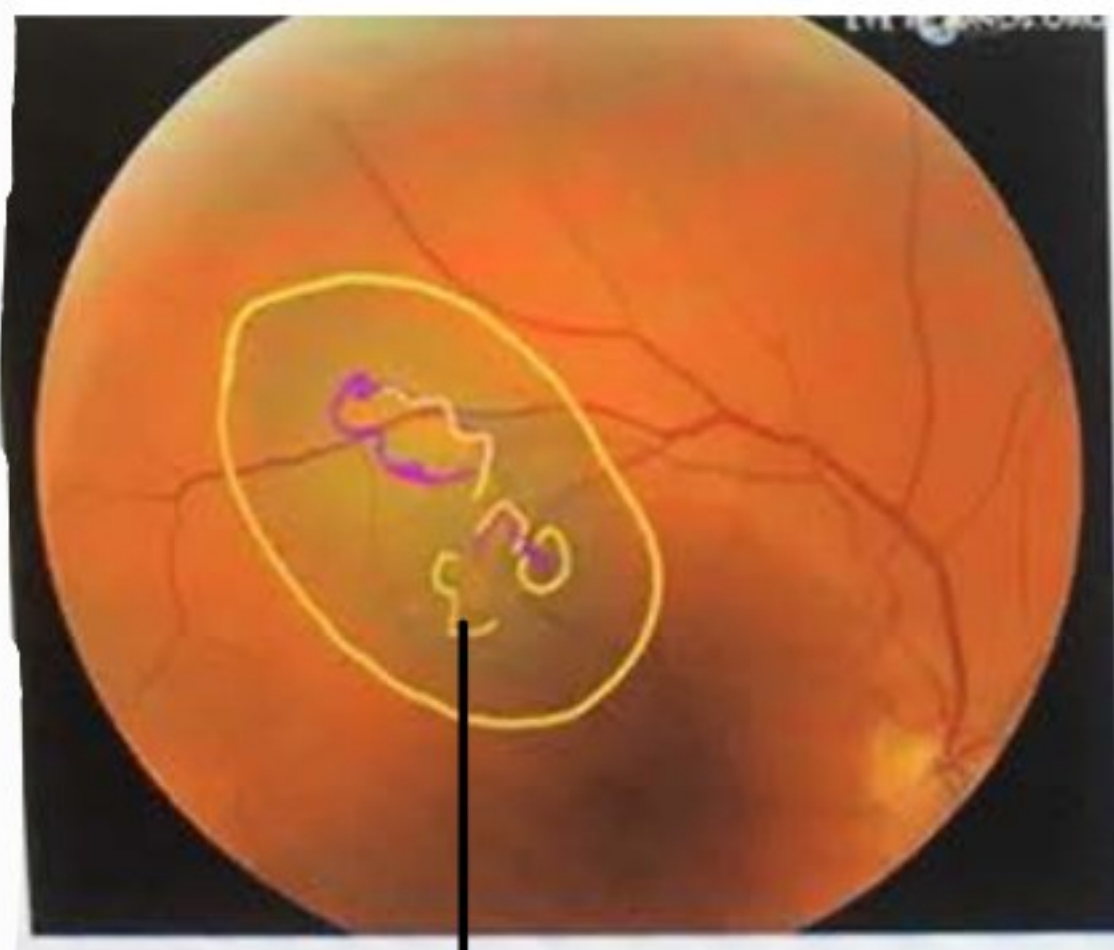
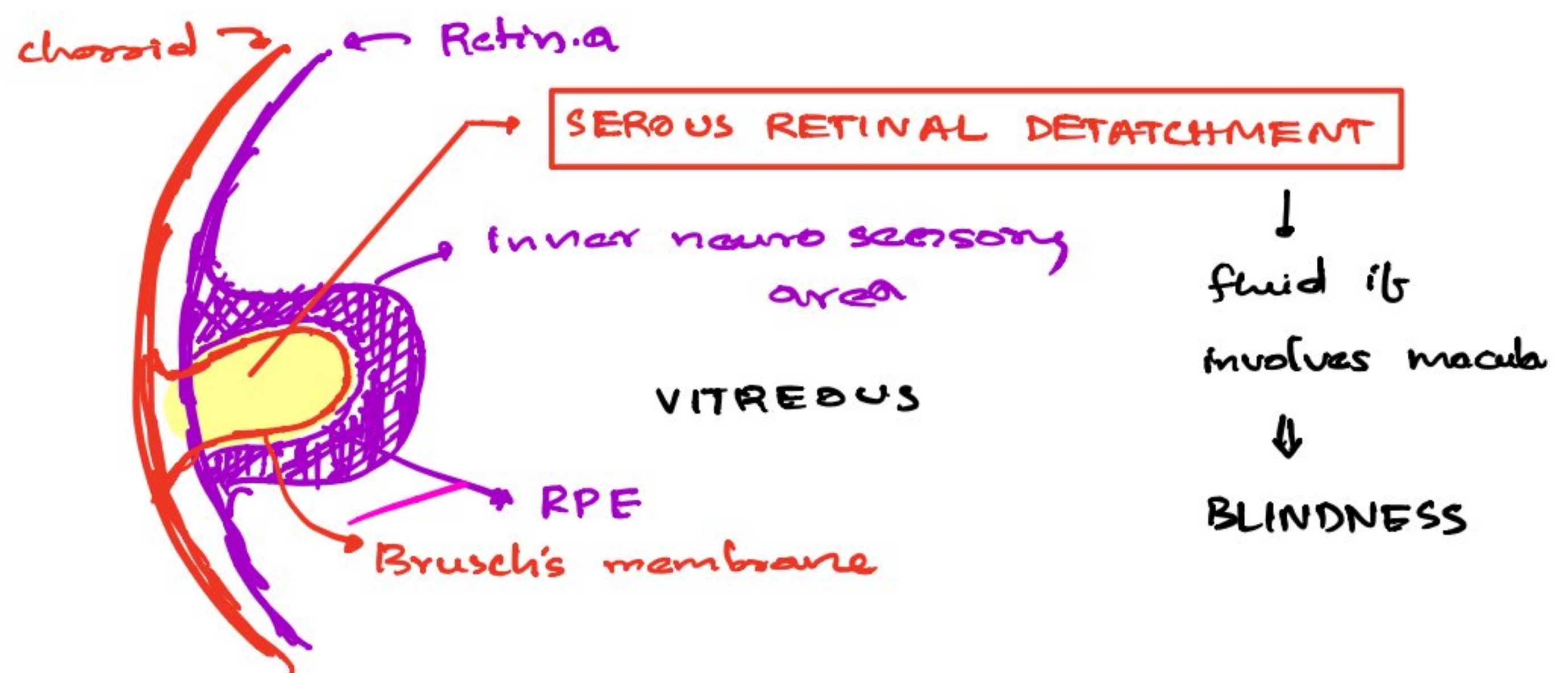
(MC) 1° malignant intraocular tumour in adults

• Pigments: i) melanoma +/-

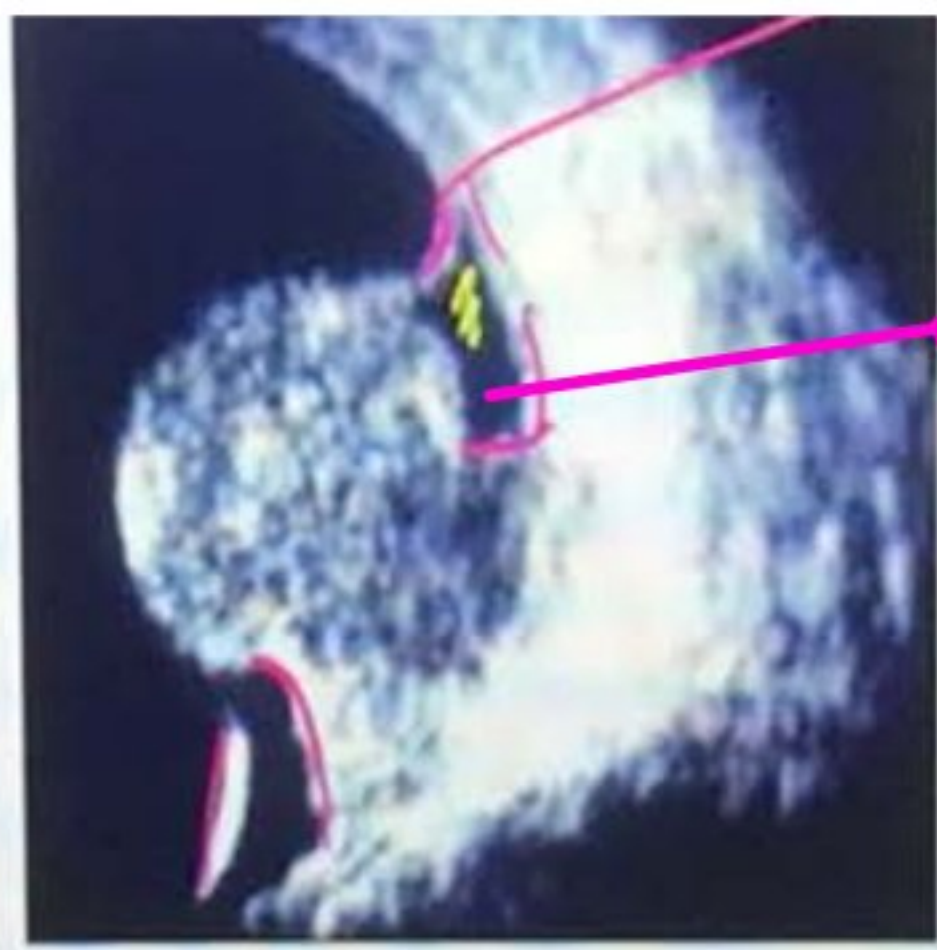
ii) LIPOFUSCIN → Pathognomic



Choroidal melanoma



LIPOFUSCIN



B-scan. → for size

collar stud app/ mushroom shape tumour.

• Fastly metastatic ∴ Rapid treatment required

• (MC) site of metastasis → Liver > lungs

{ Haematogenous via Vortex Vein }

① V. small tumour = close observation

② Tumour  $\tau$  Basal diameter upto 10mm } BRACHYTHERAPY  
Thickness  $\longrightarrow$  2mm }  
L. Radioactive  $I^{125}$  plaque sutured  
on to sclera over the tumour for 3 days

⊕ Focal treatment in addition to Bachytherapy

i) TTT = Transpupillary thermotherapy  
Diode laser  $\rightarrow$  IR (heat)

ii, PDT  $\rightarrow$  Photodynamic therapy

3, Large tumours : Enucleation of eyeball HPE

↓

Follow up : USG - Abdomen } to R/o  
                  &  
                  LFT         } Liver mets

CXR  
HRCT chest } to R/O lung mets

HPE → CALENDAR CLASSIFICATION

- ↳ Spindle A tumour → well diff.  
V. good Prx

spindle shape

2, Spindle B tumour → well differentiate  
good Prx



## Plump spindle

3) Mixed type → spindle + epithelioid

### \* Intermediate Prognosis

#### 4) Epithelioid type



- \* poorly differentiated
- \* Worst prognosis.

HIV Eye NEET PG '18

Q: (MC) ocular feature of HIV infection is Microcephaly

## H'maze

## Microaneurysm

Cotton wool spots  
/ soft exudates

## Opportunistic infection of the eye in HIV

b) CMV  $\rightarrow$  (MCC) 2<sup>o</sup> post. uveitis in HIV

ii) Toxoplasmosis

iii) *Pneumocystis carinii*

iv, H. Zoster

UPSC '13  
AIIMS '13

(MC) finding of HIV retinopathy

HAART → (MC) ocular adverse effect → IMMUNE RECOVERY UVEITIS

{Ant. non granulomatous uveitis}

• d/t paradoxical / exaggerated reactiv. of the immune system and an increase in intraocular inflammation

4/4 complication :  
i) Cataract  
ii) Cystoid macular edema  
iii) Glaucoma

### Drug induced Uveitis

- 1) Cidofovir
- 2) RIFABUTIN
- 3) Bisphosphonates
- 4) Oral Moxifloxacin

### UVEAL EFFUSION SYNDROME

- B/L
  - Idiopathic (rare)
  - affects middle aged **HYPERMETROPIC MEN** a/w nanophthalmos
- cause: Impairment of (N) fluid drainage from choroid via sclera / vortex veins
- signs :  
i) No inflammation  
ii) Ciliochoroidal detachment  $\xrightarrow{f/b}$  exudative retinal detachment  
iii) Resolution → RPE show "Leopard spot" mottling

Rx: Full thickness sclerectomy.

ENDOPHTHALMITIS : Suppurative inflammation of layers of eye of infectious etiology onset as → purulent and uveitis (iridocyclitis)  
or  
purulent post uveitis (choroiditis)

$f/b$  ↓  
involving retina & vitreous.

Modes of infection: Early Endophthalmitis < 4wk after cataract surgery

- i) Perforating injuries ← (MC) Staph. Epidermidis
- ii) Endogenous or metastatic → Through blood stream from some infected focus such as carries tooth, puerperal sepsis.
- iii) 2° infection from surrounding suture ← MCC Staph aureus.

## Symptoms:

- Severe ocular pain
- Redness
- Lacrimation
- Photophobia
- loss of vision

Prophylaxis: To prevent surgery induced

Endophthalmitis

i) 5% povidone I<sub>2</sub> eye drops x 3min  
before starting Sx

ii) Intracameral Antibiotic (AC) = MOXIFLOXACIN  
after surgery

## Signs:

- 1) Lids → Red, swollen
- 2) Conjunctiva → congestion, chemosis
- 3) Cornea → ring infiltration
- 4) Ant. chamber → Hypopyon
- 5) Pupil → Yellow reflex +ve (Amaurotic Reflex)
- 6) IOP ↑sed, later decreased d/t destruction of ciliary process.

Rx :: Vitrectomy / intravitreal antibiotics.

## LATE ONSET ENDOPHTHALMITIS:

Intraocular infection of vitreous cavity after prev. intraocular surgery [ $>6$ wks]

## Etiology:

Causative organism: Fungus: Candida Albicans

(MC) bacteria: Propionibacterium Acne

Presentation: May occur P. acne → from 3 months to 2 years

CANDIDA ENDOPHTHALMITIS → within 3 months

- Cottol ball in vitreous
- String of pearl appearance

## CASE SCENARIO

A 65 yr old male presents with a painful decrease in vision of his R eye for the past 3 days. He underwent right Phacoemulsification with PC-10Z insertion 6 months ago. Examination revealed a mild granulomatous anterior uveitis with minimal hypopyon and mild vitritis. There is a white plaque seen on the P lens capsule. Which one of the following is the most appropriate therapy for this condition?

sol: Vitrectomy + posterior capsulectomy and intravitreal injection and vancomycin.

H/o pt. Have variable decreased vision with photophobia,  
• gradual onset of symptoms

• Infection may be painless

Clinical sign ∴ Vitritis with/without hyphema > 6wk after surgery.

- KP's
- White plaque on post. lens capsule.

Prognosis and management : intravitreal tap → Cs → PCR analysis

NOT RESPONDING > 48hr

↓

⊕ Vitrectomy with explanation of IOL & capsular bag.

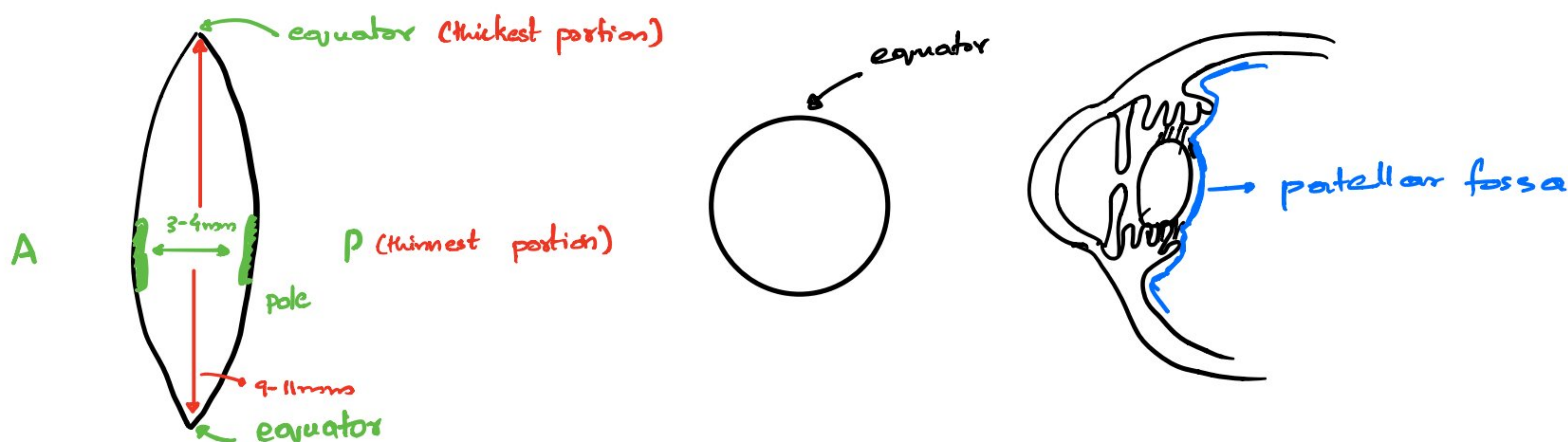
↗

Bacterial: Intravitreal Vanco + Cephalosporin + **DEXA**

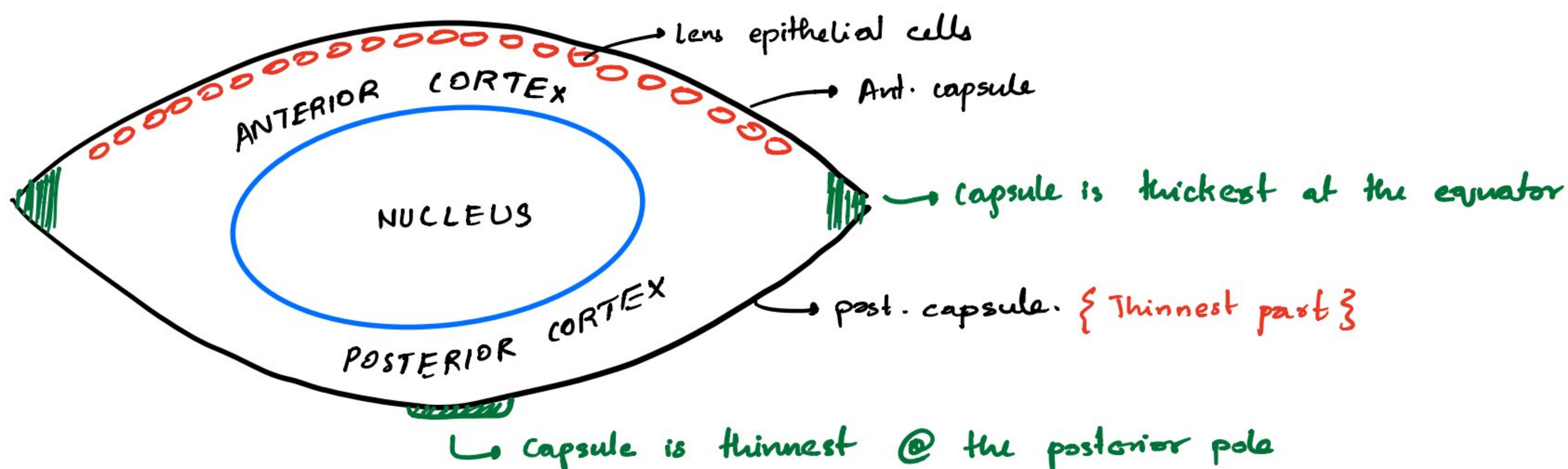
Fungal: Intravitreal Amphotericin B

LENS ∴

• Shape = BICONVEX



Equatorial cells is the most active cells of lens.  
a.k.a bladder cells of Wedl.



## Layers

- 1) Ant. capsule
- 2) Ant. epithelium
- 3) Lens fibres

NUCLEAR FIBRES

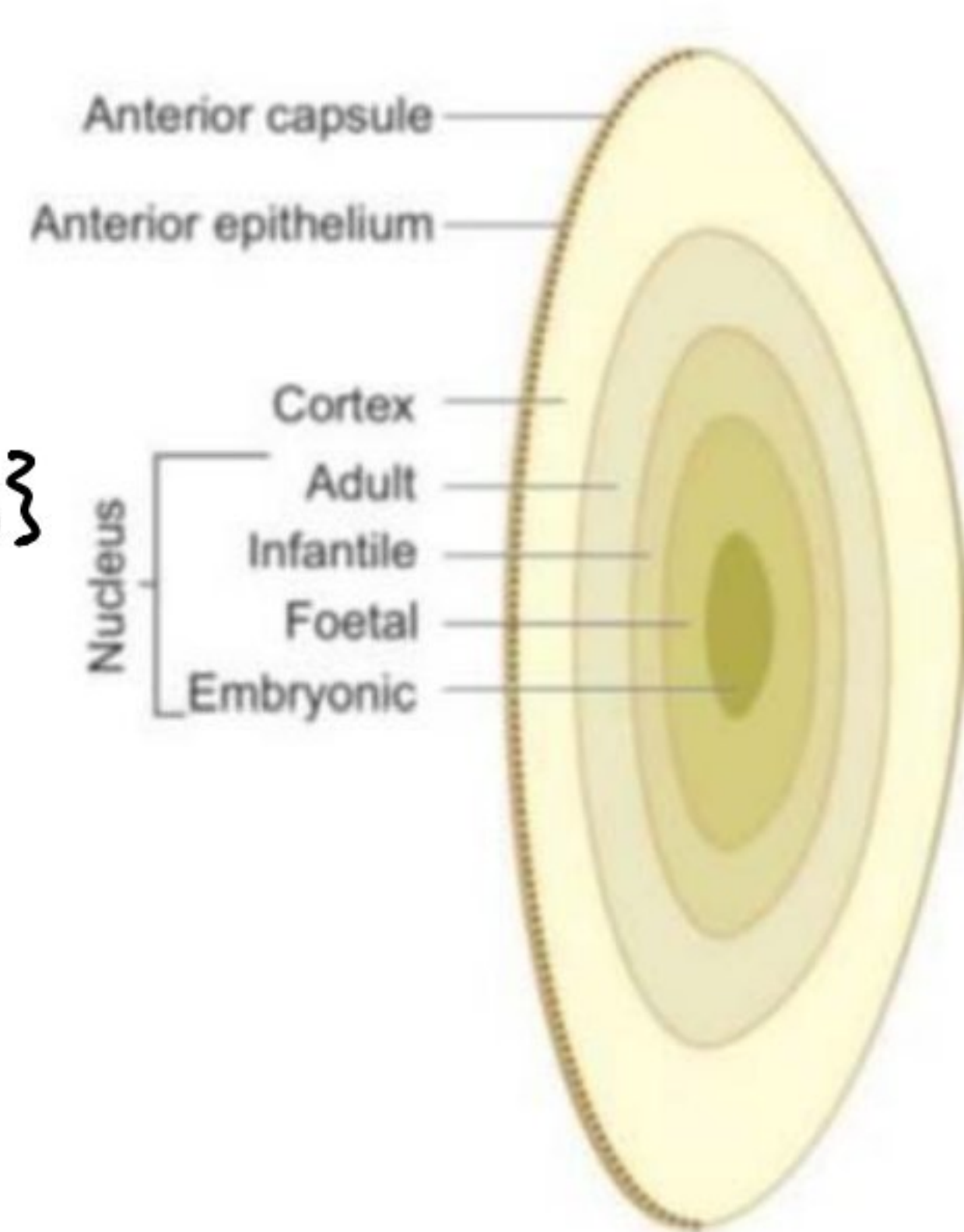
Cortical (central)

- Embryonic  $\{ 1m - 3m \text{ of gest.} \}$
- Fetal  $\{ 3m - \text{birth} \}$
- Infantile  $\{ \text{Birth} - \text{puberty} \}$
- Adult  $\{ > \text{Puberty} \}$

Overall youngest fibres

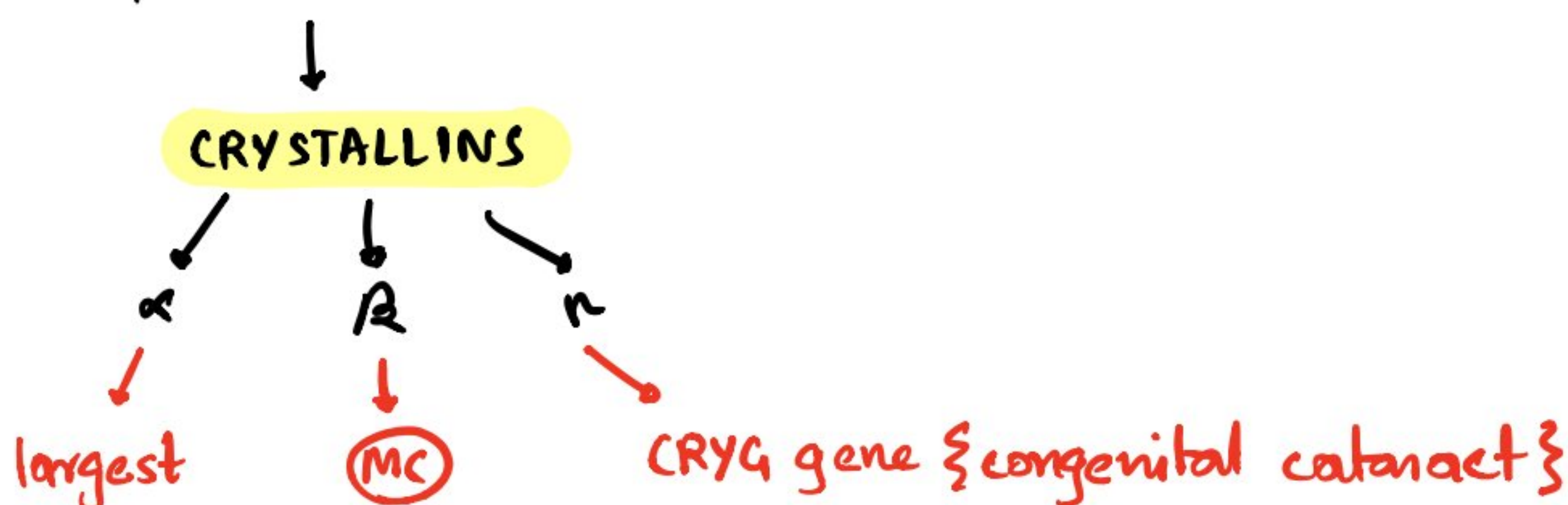
x no post. epithelium

- 4) Post. capsule



⊕ Radius of curvature → Anterior lens = 10mm  
→ Posterior lens = 6mm (more curved)

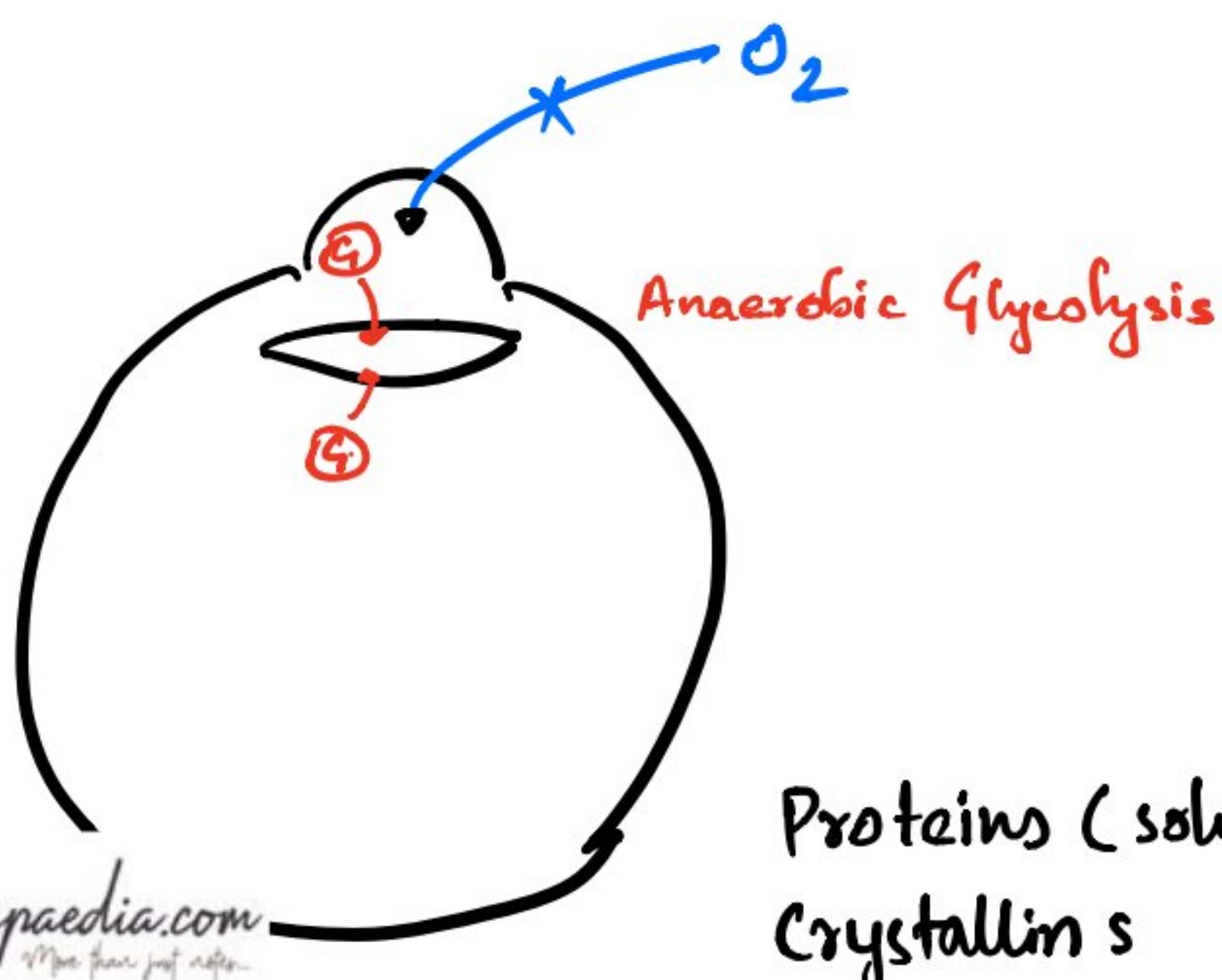
- 66% of lens made up of water
- 34% " " " " " proteins (soluble)



## Factors responsible for transparency of lens

- 1) Crystallins
- 2) Lamellar arrangement of lens
- 3) Avascularity
- 4) Free Radical scavengers of lens → Eg.:
  - Glutathione → G6PD deficiency / superoxide desmutase
  - Vit C { myonisol transp. Vit. C }
  - Vit E / Glutathione peroxidase
  - Carotenoid { Vit A. not present in lens }

## Nutrition of lens : from Aqueous humour



### MIP-26 ⇒ Major Intrinsic Protein

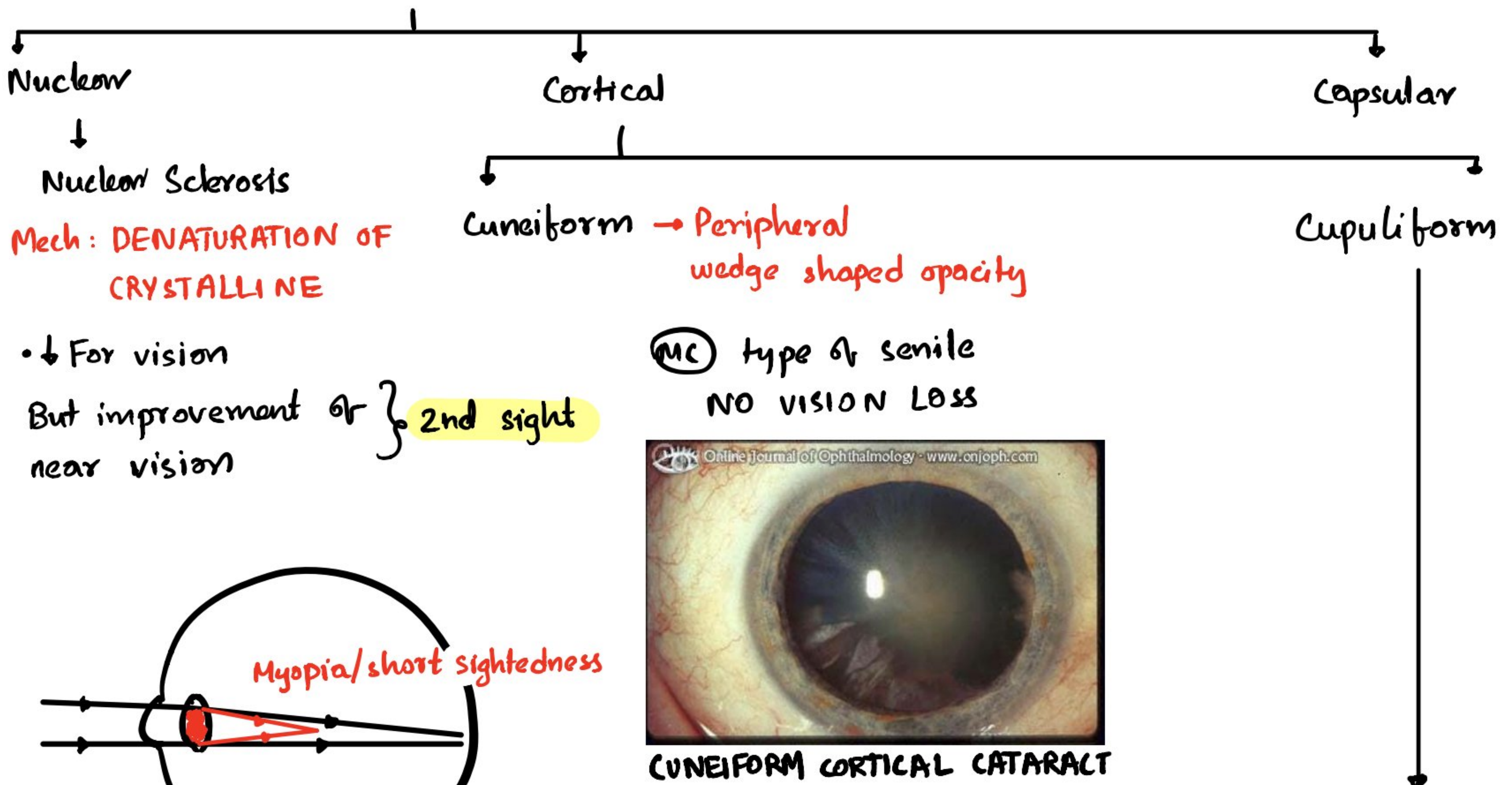
- Regulates H<sub>2</sub>O entry into the lens a.k.a Aquaporin 0
- Its defect L/E cortical cataract (↑sed water entry)

Insoluble ~ Opacification of lens  
Albuminoid CATARACT

CATARACT : Any opacity in the lens

cause : Age  $\Rightarrow$  • Senile Cataract  
• Gradual painless vision loss

### Types of Cataract



Lens Changes : i) Hardening  
ii) Discolouration

Yellow → Brown → Black

d/t **Urochrome pigments**

d/t 3-hydroxykynurenin  
a derivative of TRYPTOPHAN

Cataracta  
Brunesens

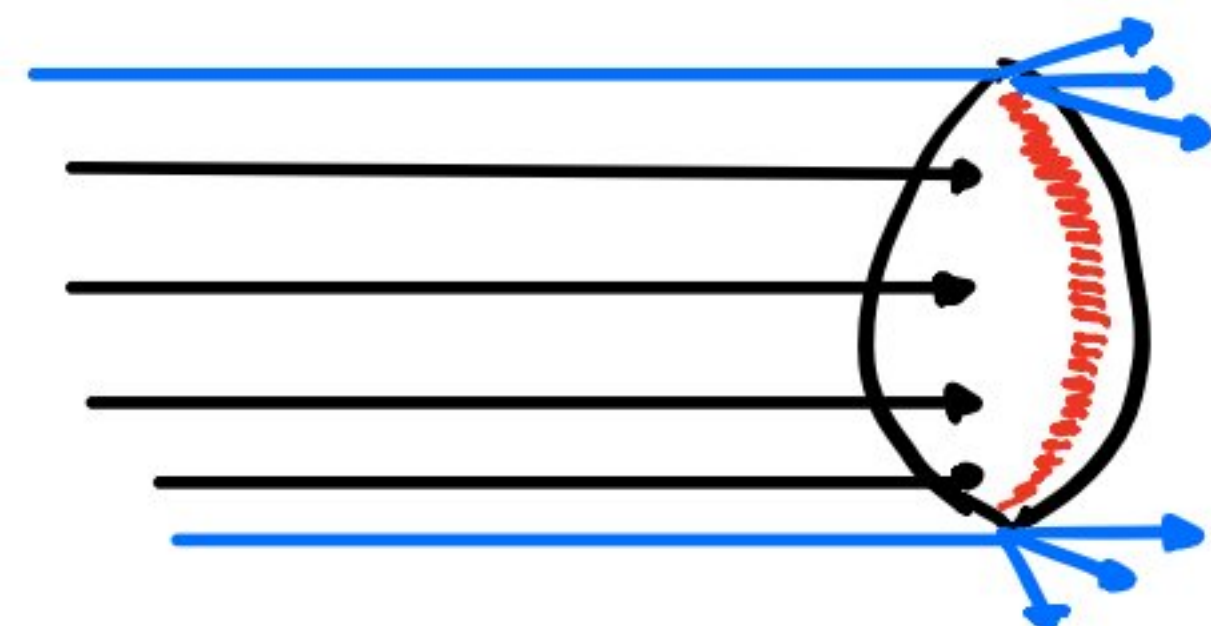
Cataracta  
Nigra

• Cataract is at post.  
surface / post. subcapsu  
lar



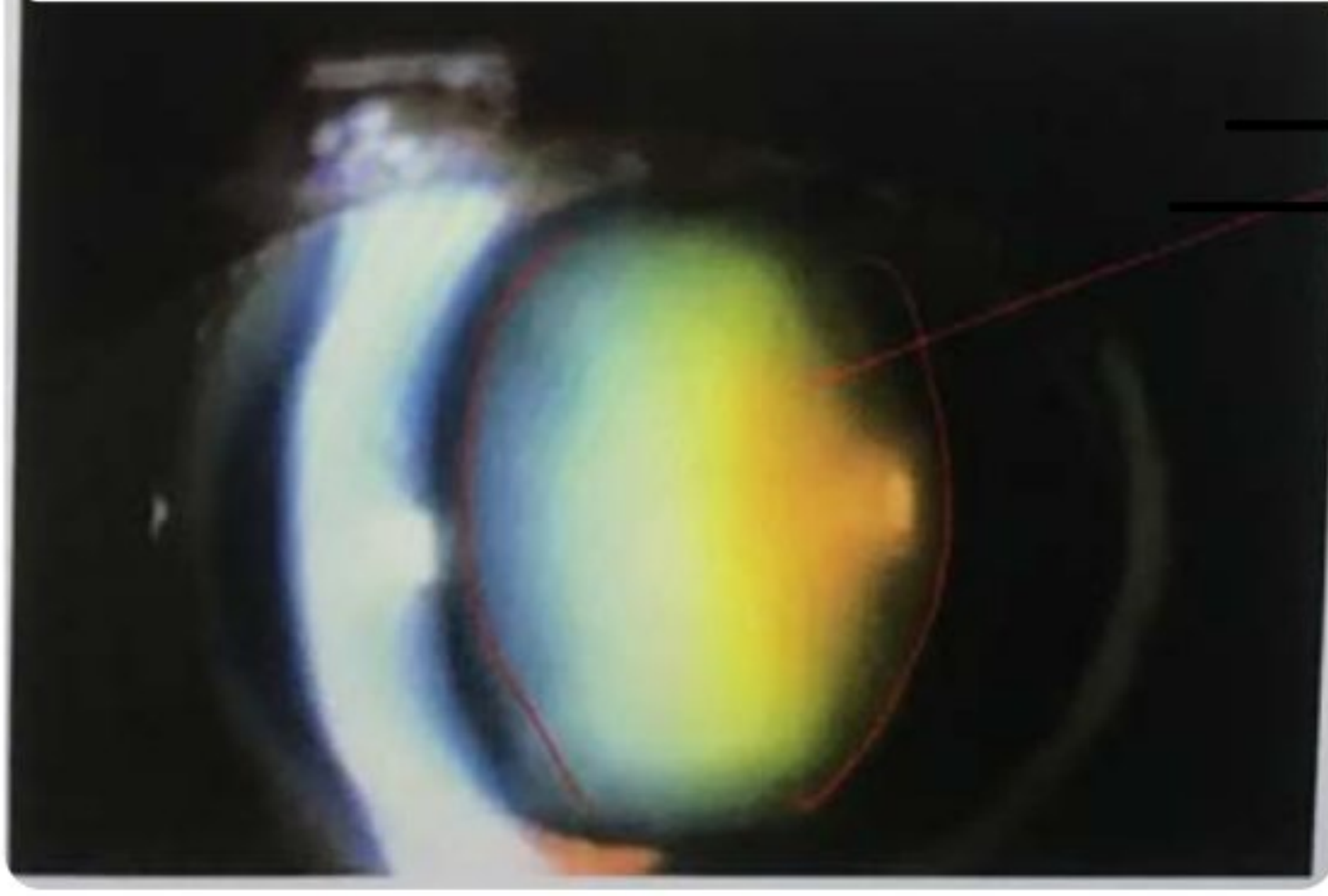
• Both far and near vision loss  
• Glare at night while driving

Dilated  
pupil



R.F. : i) Aging  
ii) Sunlight (U.V)  
iii) Def. of Vit. A, C, E  
iv) Smoking  
v) Type 2 DM

## NUCLEAR SCLEROSIS



### Capsular Cataract

- 1) Anterior Capsular
- ii) Anterior Subcapsular



- cause
- 1) Chlorpromazine
  - 2) Amiodarone
  - 3) Glaucoma fleck
  - 4) Pilocarpine {M3 Agonist}
  - 5) Electric shock
  - 6) Phenothiazines
- ↓
- LONG TERM USE

- ii) Anterior polar cataract



- causes
- i) Perforating corneal ulcer / injury
  - ii) Pilocarpine

### 2) Posterior Capsular Cataract :-

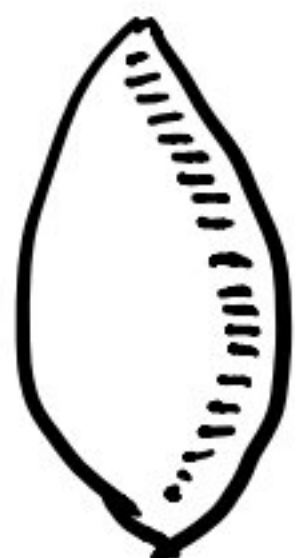
- i) Posterior Subcapsular cataract.

- causes
- a) Complicated cataract
  - b) Cupuliform Cataract.
  - c) Bisulfan
  - d) **STERIODS** {systemic > topical}
  - e) Chloroquine.
  - f) Radiation cataract

1. R/ Heat cataract  
a.k.a glass blowers cataract

Ionizing Radiation  
 $\alpha$ ,  $\beta$ ,  $\gamma$  Xrays

PSC

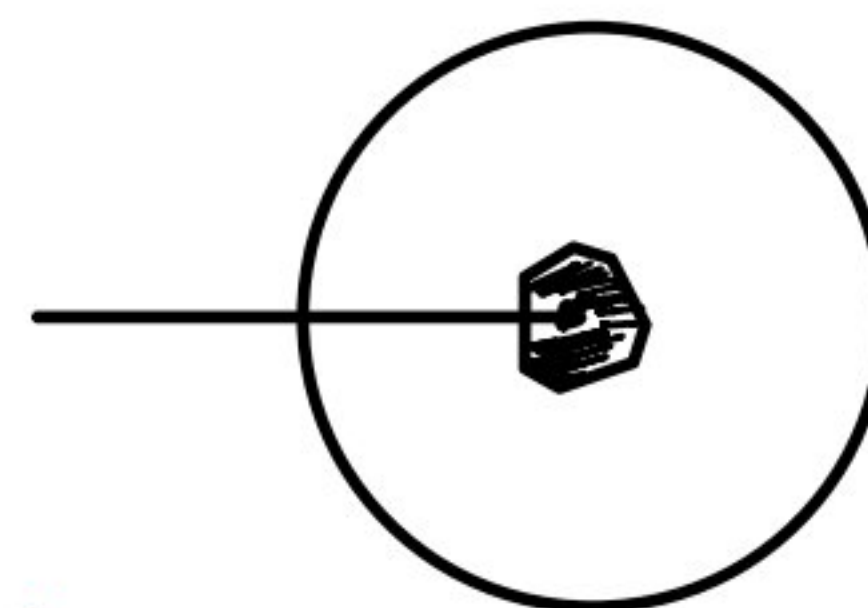


symptom : GLARE

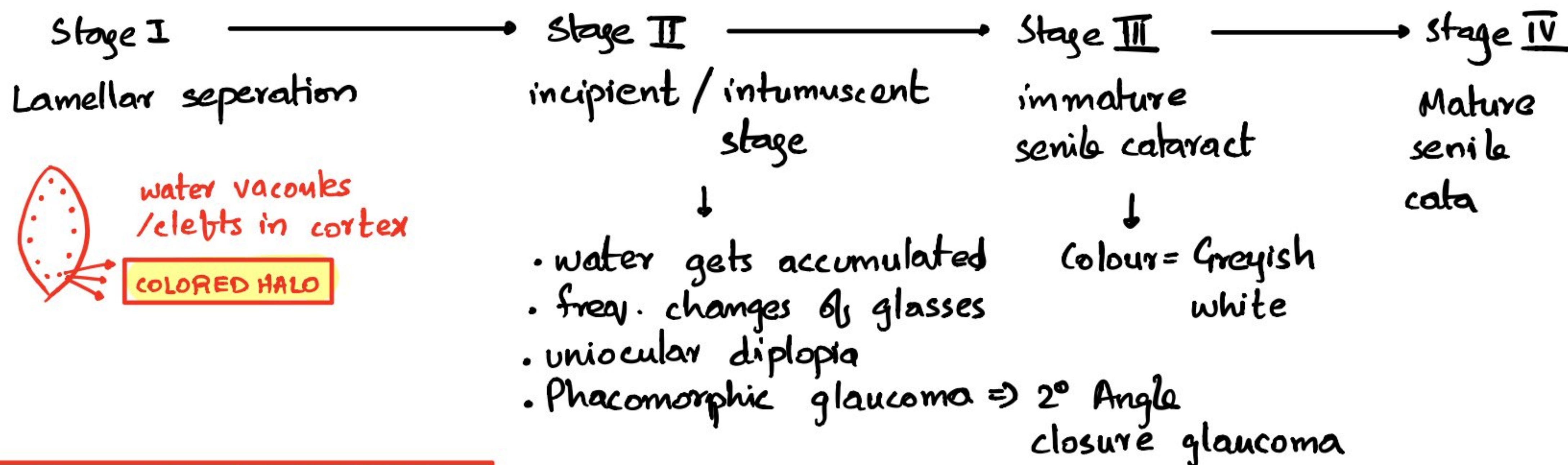
## ii) Posterior polar Cataract



- Congenital
- whorled or "onion ring app"
- 90% MITTENDORF'S DOT
- posterior capsule rupture is v. high d/t surgery

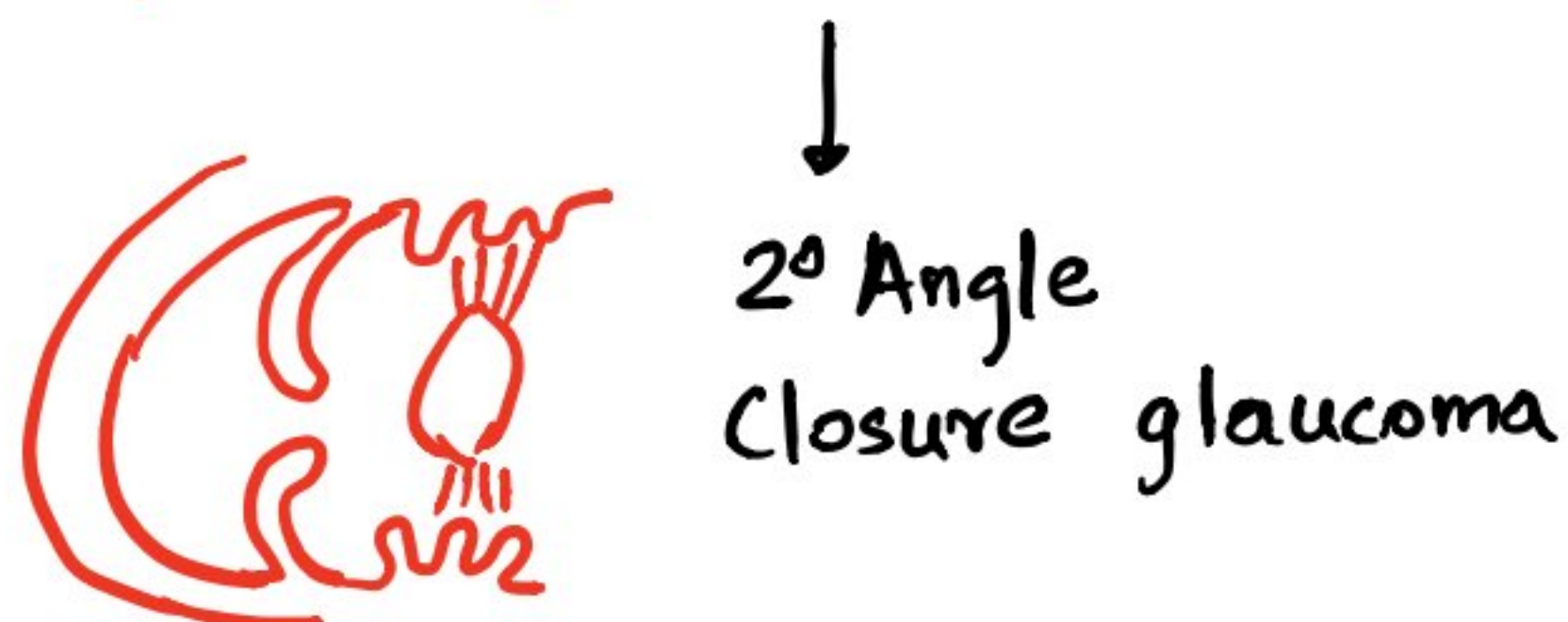


## Stages of Senile Cataract:



## INTUMESCENT CATARACT

swollen cataractous lens  
d/t rapid entry of  $H_2O$  → lens



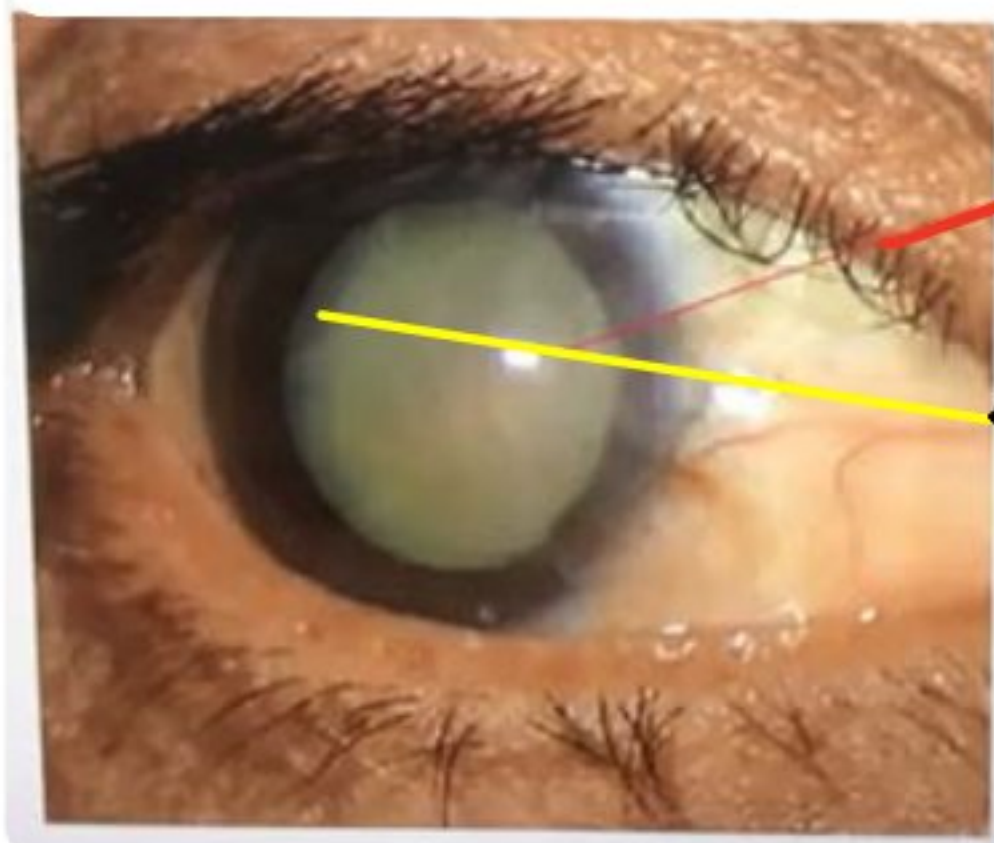
## PHACOMORPHIC GLAUCOMA

- AC depth shallow

• More commonly causes glaucoma



## IMMATURE CATARACT



Mix of opaque & clear lens

Iris shadow ⊕

## MATURE CATARACT



- Total Opaque
- Iris shadow ⊕
- Distant direct ophthalmoscopy
- ↓
- No fundal glow

## Stage II:

### HYPERMATURE CATARACT

1) Sclerotic → Calcific degeneration



weakening  
of zonular attachment  
on the capsule

⇒ **LENS  
DISLOCATION**

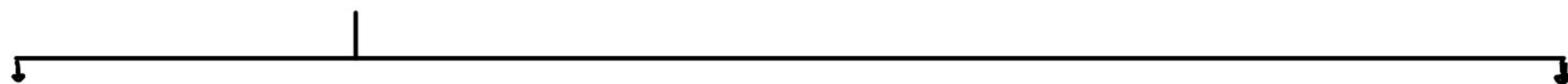
2) Morgagnian Cataract → Liquefaction degeneration

- Cortex → Milky & white fluid
- Nucleus → Sink in the cortex
- Capsule → wrinkled & leaky

### Unilateral Diplopia

- 1) Astigmatism
- 2) Subluxated lens
- 3) Polycoria { more than one pupil }
- 4) Incipient cataract

### Hyper mature senile cataract:



Sclerotic → small fibrotic lens  
subluxation / dislocation  
phacolytic glaucoma

Phacomorphic  
Glaucoma

>

Phacolytic  
Glaucoma



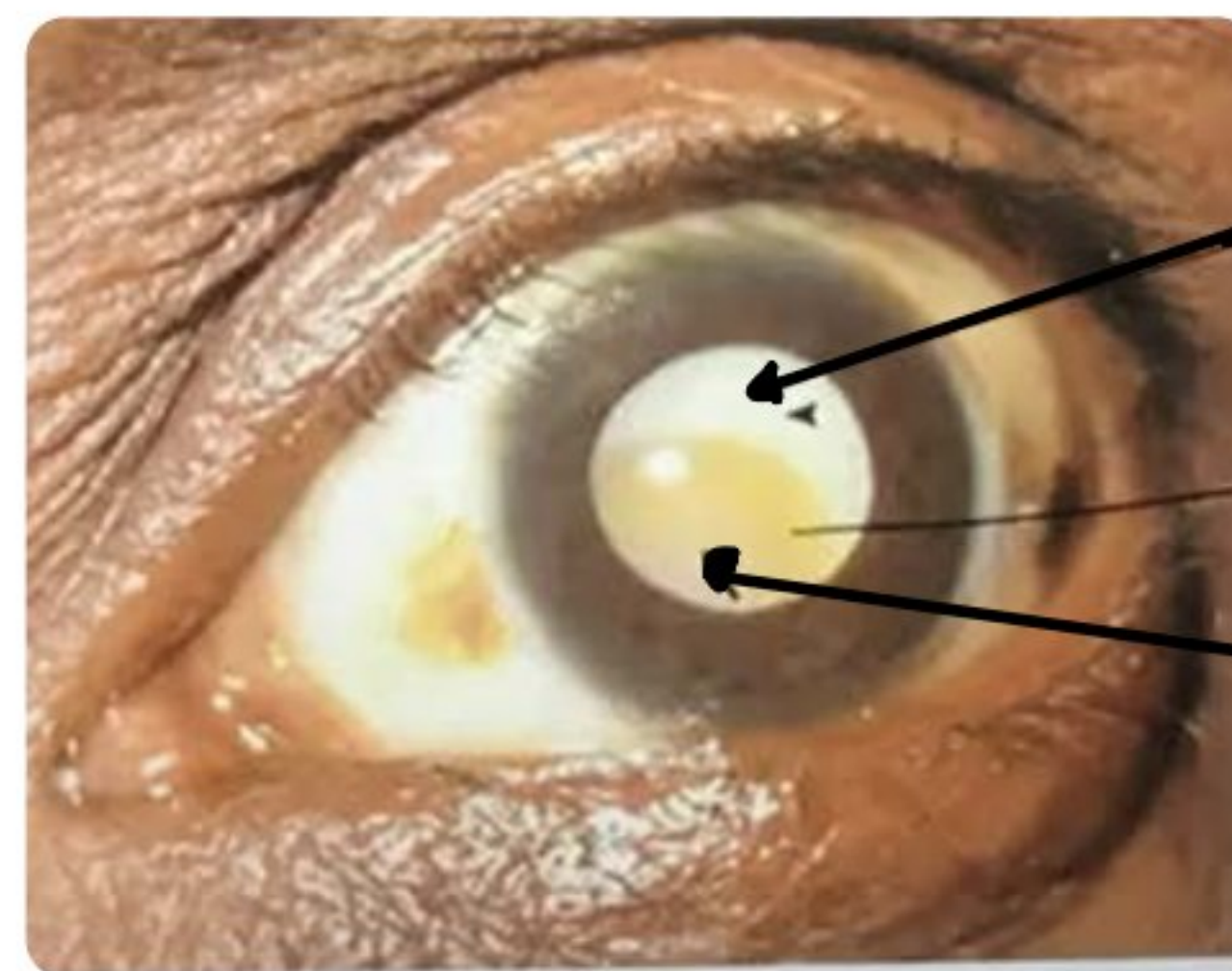
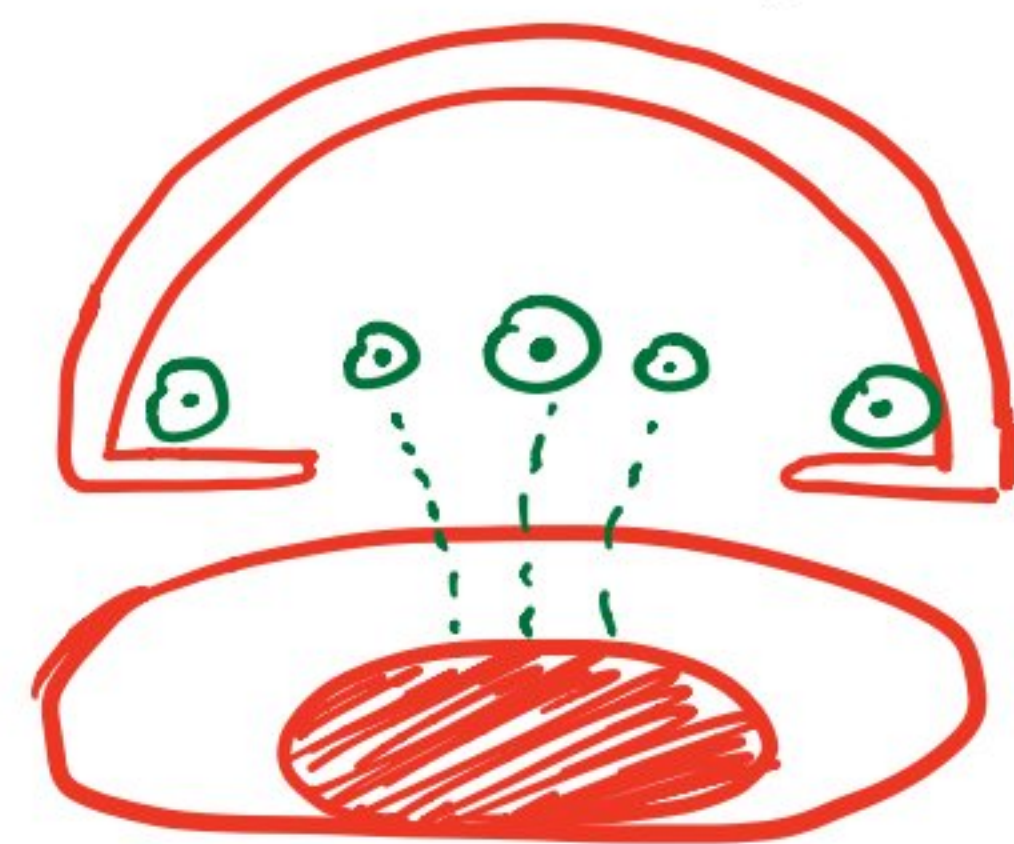
Phacolytic  
glaucoma

} AC depth  $\downarrow$  /  $\uparrow$

Lens protein released into  
ant. chamber & block the  
Trabecular Meshwork

} 2° open angle  
glaucoma

Morgagnian



Morgagnian Cataract

milky  
white  
cortex

Dense  
Brown  
cataract

## SYMPTOMS OF CATARACT :

- 1) Slow progressive **PAINLESS** diminution of vision  
& no nerve fibres in lens

If painful, red eye in the setting of cataract  $\Rightarrow$  Lens Induced Glaucoma

- 2) Second sight  $\Rightarrow$  Nuclear Sclerosis  
3) Glare  $\Rightarrow$  PSC  
4) Better vision in dim light } centrally located cataract  
or  
Blinded in bright light }  
[HEMARALLOPIA] }  
• PSC  
• N.S  
• Cupuliform cataract

Q. Hemaralopia is s/i in all except ?

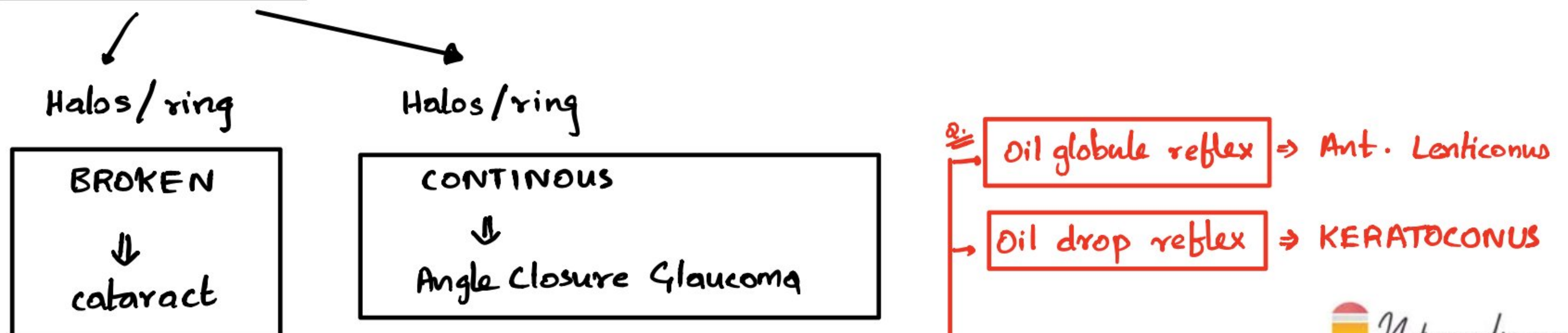
- a) PSC  
b) Cupuliform Cataract  
c) N.S  
d) Cuneiform Cataract

- v) Polyopia } Early stages of cataract  
vi) Coloured Halo }

COLOURED HALO  $\Rightarrow$  D/D

- 1) Cataract  
2) A/c Congestive Glaucoma  
3) Mucopurulent Conjunctivitis [Bacterial Conjunctivitis]

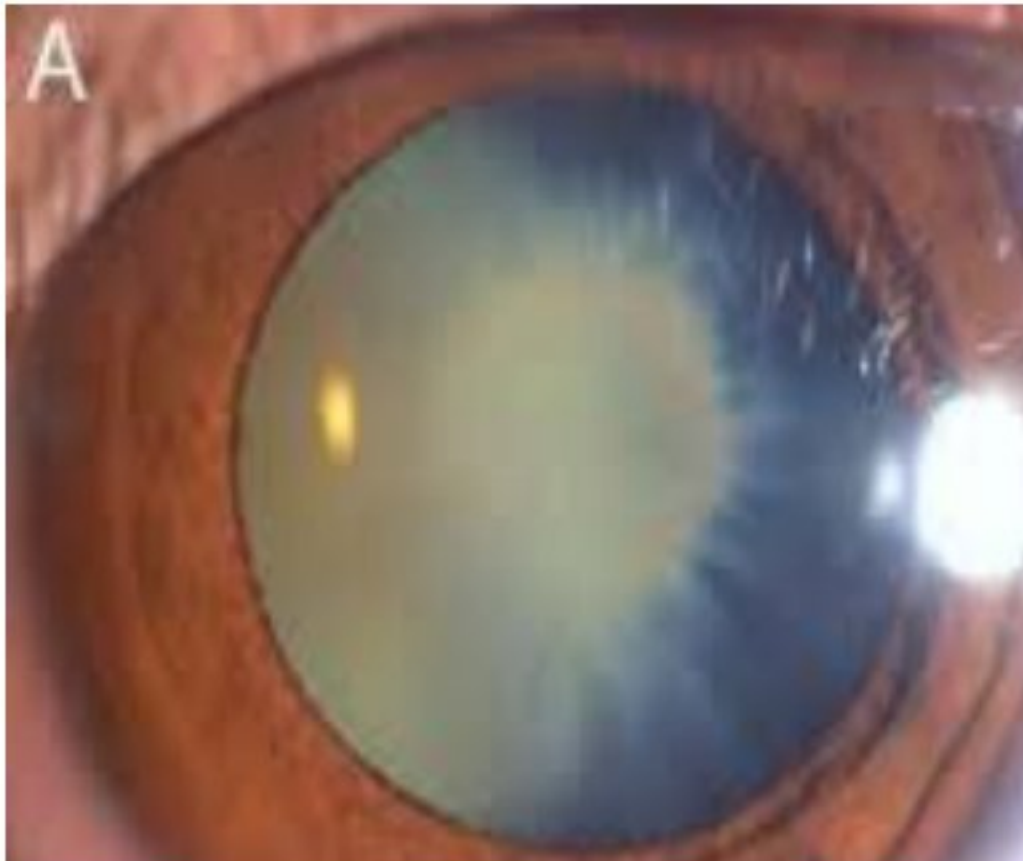
**FINCHAM'S TEST:** {different cataract vs ACG causes of Halos?}



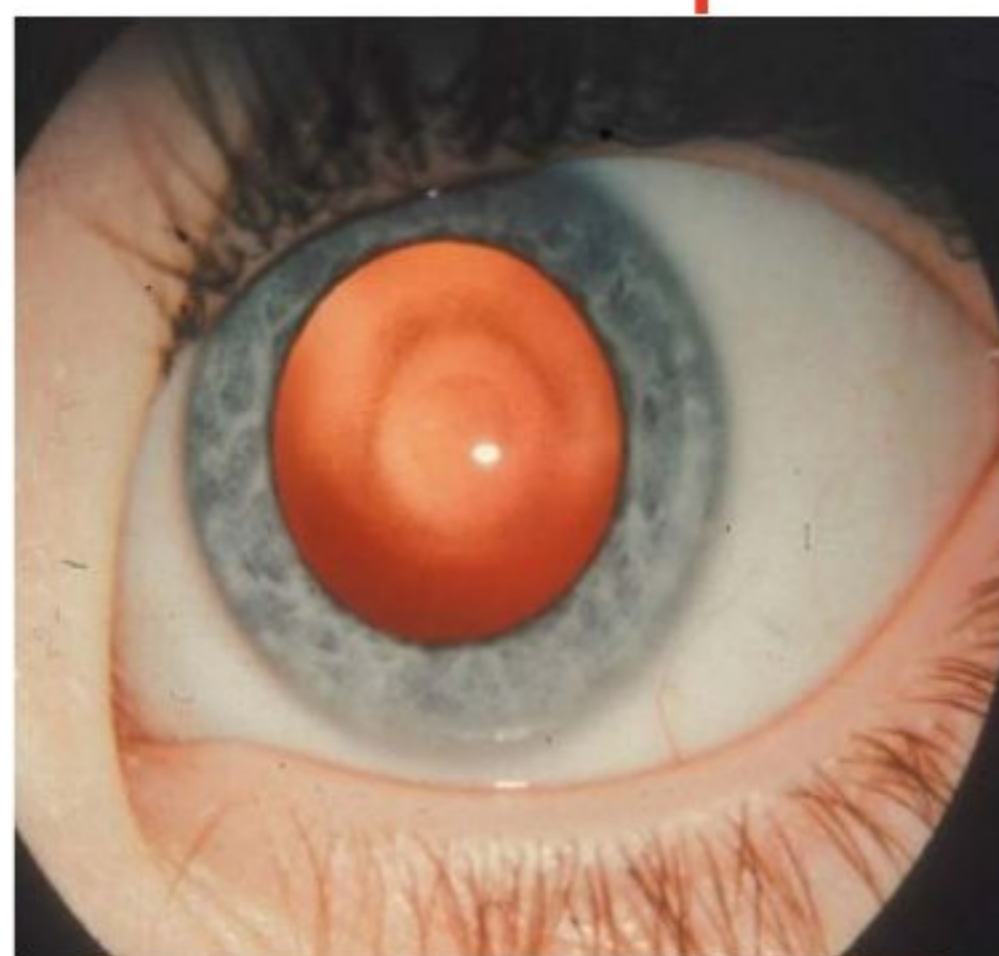
**ROSETTE CATARACT** **SUNFLOWER CATARACT**



Blunt Trauma

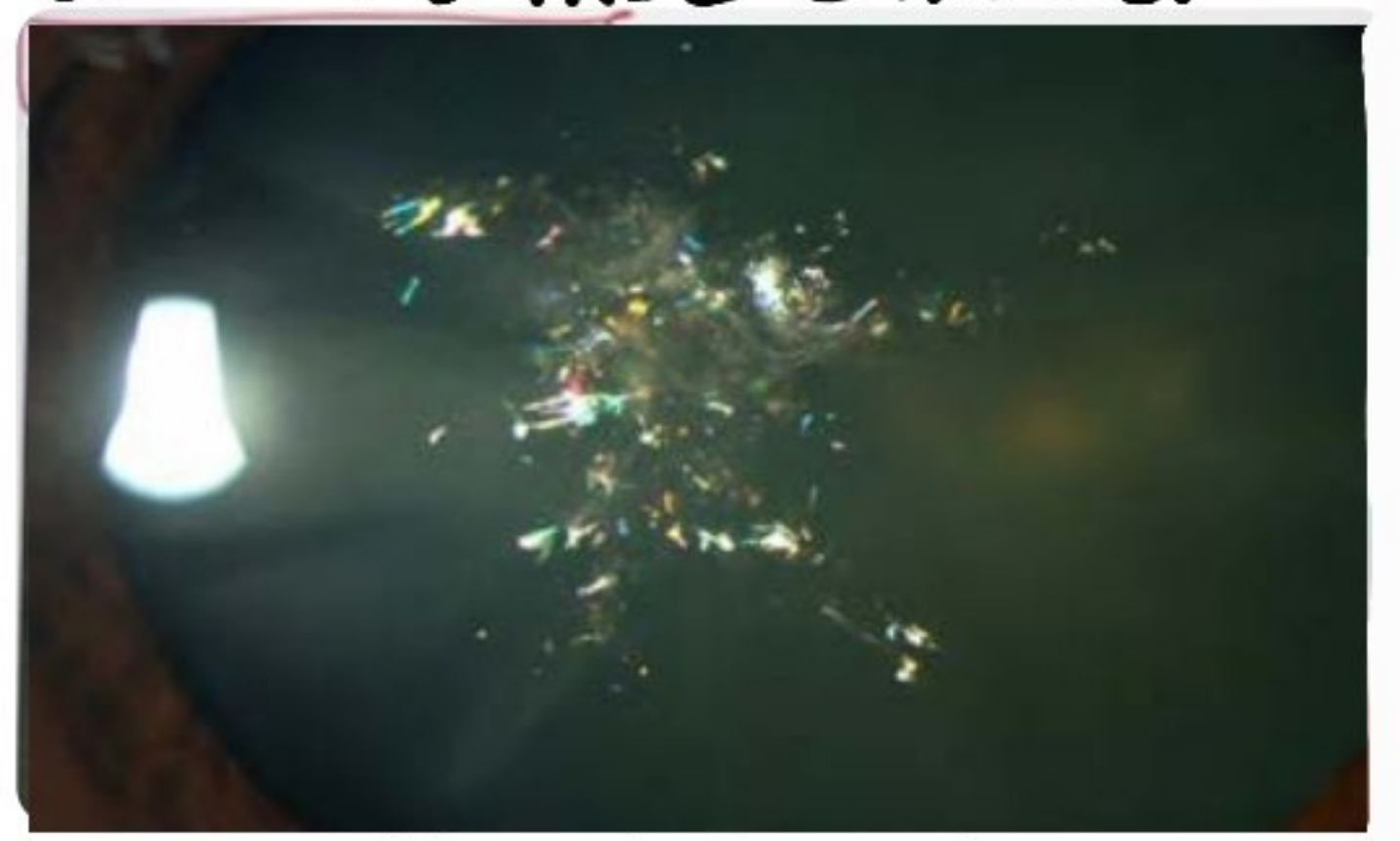


Wilsos Disease and  
Chalcosis



Galactosemia

**CHRISTMAS TREE CATARACT**



Myotonic Dystrophy

## Congenital Cataract:

• Hereditary

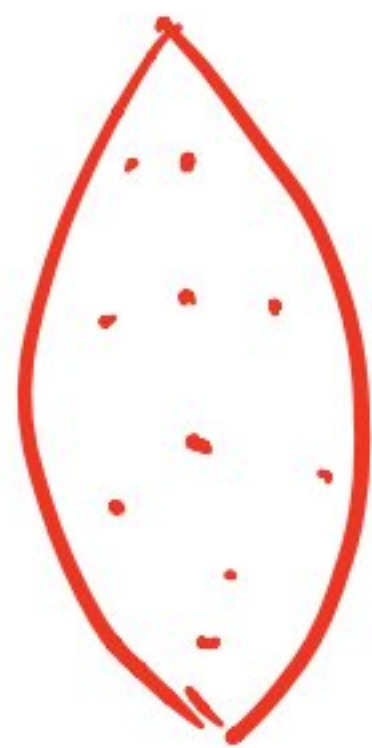
• AD

• CRYG gene defect

• (MC) type → Blue dot cataract / Punctate Cataract

• Vision (N)

• Non progressive



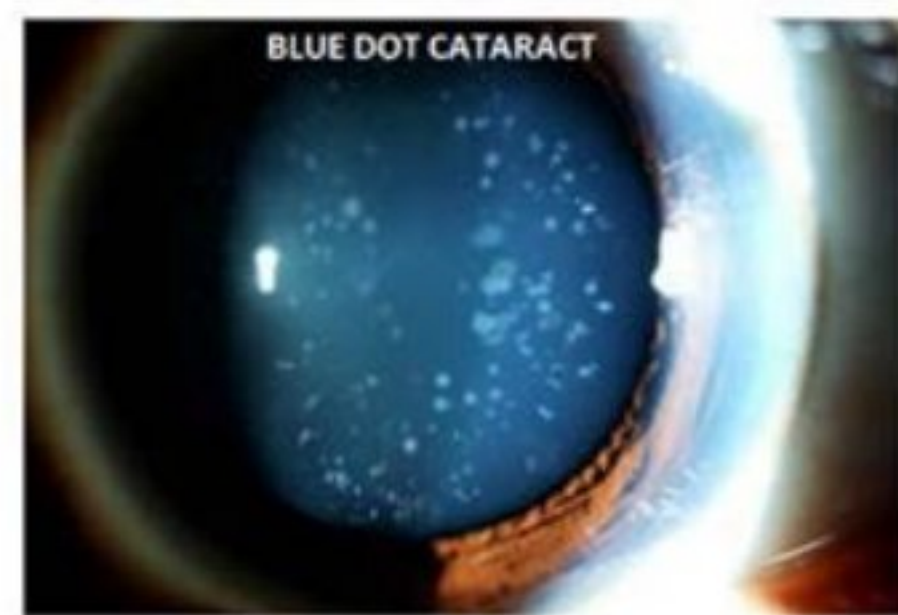
shield cornea

↓  
VKC

a.k.a

CATARACTA

PUNCTATA

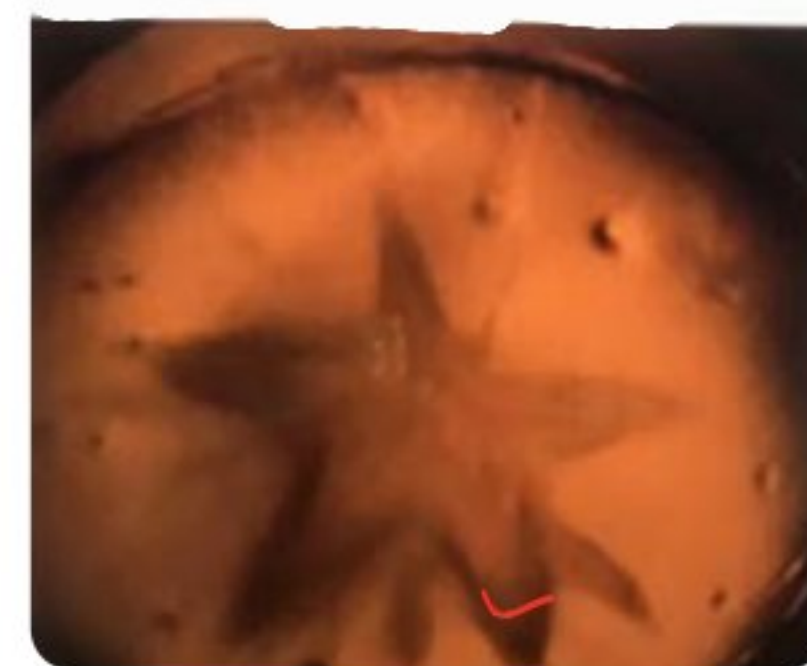


SHIELD CATARACT



Atopic Dermatitis

STELLATE CATARACT



MD and Downs Syndrome

SNOWFLAKE CATARACT

↳ uncontrolled Type 1 DM

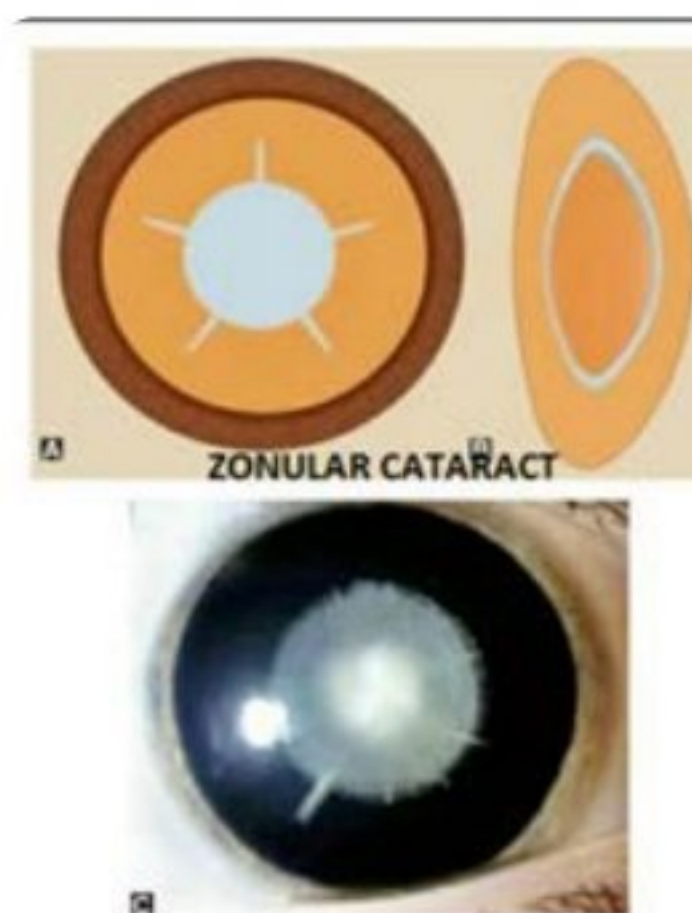
Nuclear sclerosis ← Type 2 DM

→ Zonular Cataract :

• (MC) type with loss of vision



↳ spokes & Riders pattern → pathognomic  
CART WHEEL APPEARANCE / STEERING WHEEL APP.



## Causes of Congenital Cataract :

a) Maternal malnutrition → Vitamin D

b) Infantile malnutrition → Zonular cataract

c) Maternal Infection, e.g: RUBELLA / TOXOPLASMOSIS / CYTOMEGALOVIRUS

d) Deficient oxygenation eg: PLACENTAL HEMORRAGE

e) MATERNAL DRUG INGESTION - Thalidomide, corticosteroid &

f) CATARACTA PULVERULENTA → affect the embryonic nucleus

Down Syndrome → (MC) cataract is Blue dot cataracts.

(MC) appearance : stellate cataract

## Indication of surgery in congenital cataract :

1) Centrally located cataract

2) Size ≥ 3mm in diameter & pupil size = 3mm

3) U/L cataract  $\rightarrow$  d/t Diplopia

## Timing of cataract surgery of congenital cataract

AIIMS '19 //

1-4 months of age

, As early as possible  $\neq$  4 months

FOVEAL FIXATION matures by 4 month

Sensory Deprivation Amblyopia :

- $\downarrow$
- i) Squint
  - ii) Nystagmus
- } Poor visual prognosis

To get a crisp & clear image fovea should mature, but if image falling on fovea is Hazy  $>4$  months, it remains underdeveloped.

C/F :

- i) Leucocoria  $\rightarrow$  whitish reflex of pupil



Surgery done :

Lens aspiration  
⊕  
Posterior capsulotomy  
⊕  
Anterior vitrectomy  
⊕  
IOL implantation  
 $\Rightarrow$  {undercorrected}

$\Rightarrow$  SUTURES ARE A MUST

Diabetes Mellitus type I  $\rightarrow$  Snowflake cataract

- $\uparrow$  sorbitol & defb. aldose reductase
- frequent change of glass present in DM pt, but not in diabetic cataract



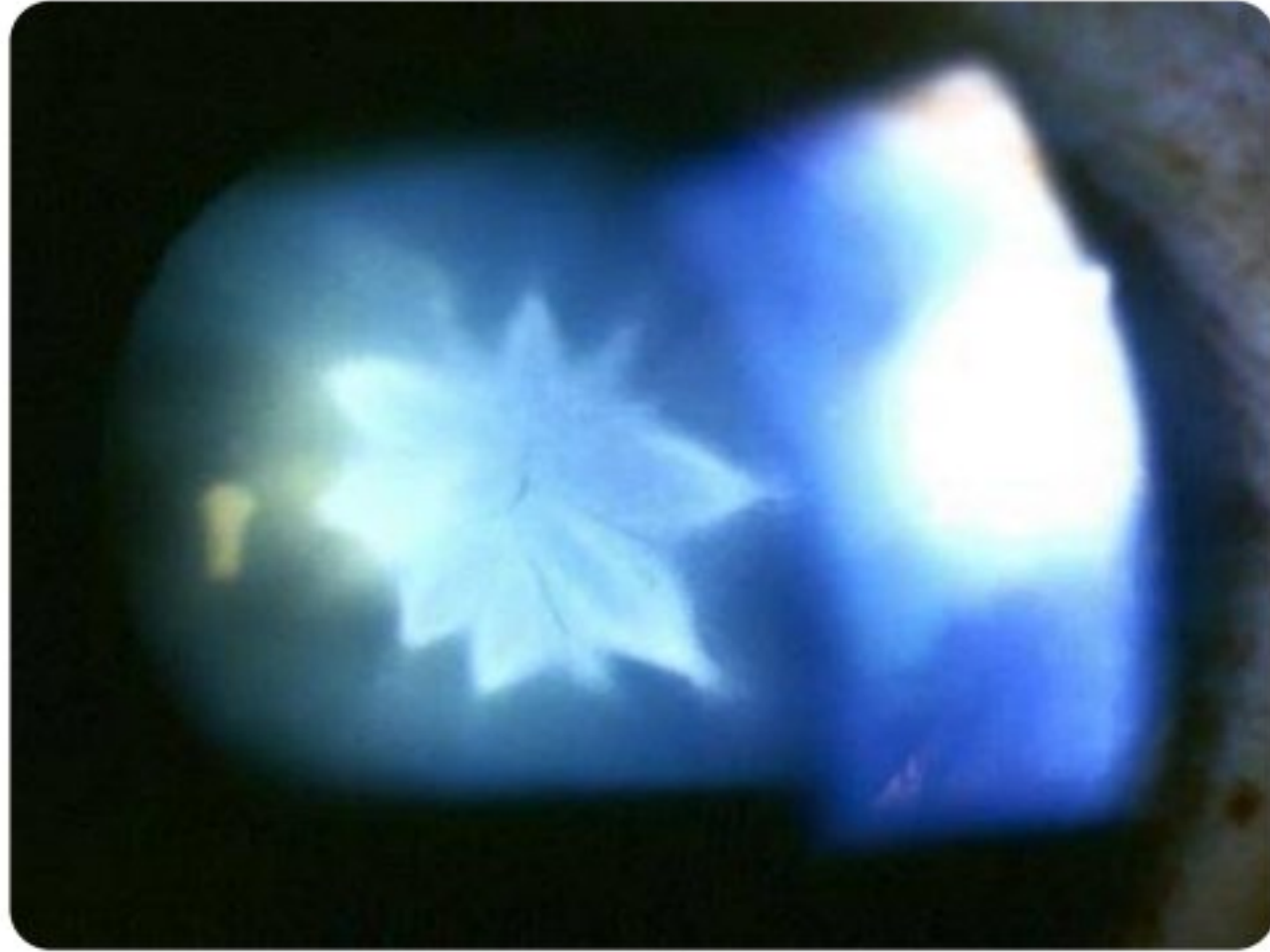
Snowflake Cataract

- Wilson's d/s  $\rightarrow$   $\uparrow$  Cu  $\rightarrow$  Sunflower Cataract
- Chalcosis
- {penetrating trauma and depositing foreign particle on to the lens}

Blunt trauma  $\rightarrow$  Rosette



Sunflower Cataract



Rosette Cataract

- (MC) Infection causing cataract  $\rightarrow$  CMV
- Pearly white cataract  $\rightarrow$  Rubella



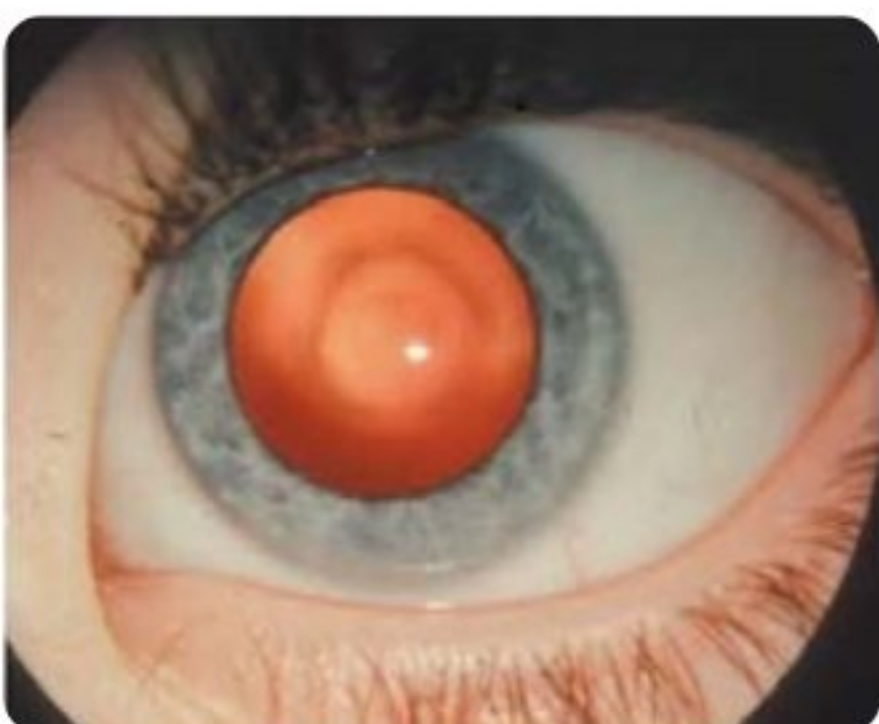
Pearly White Cataract

- (MC) chromosomal d/o causing cataract  $\rightarrow$  Down's Syndrome



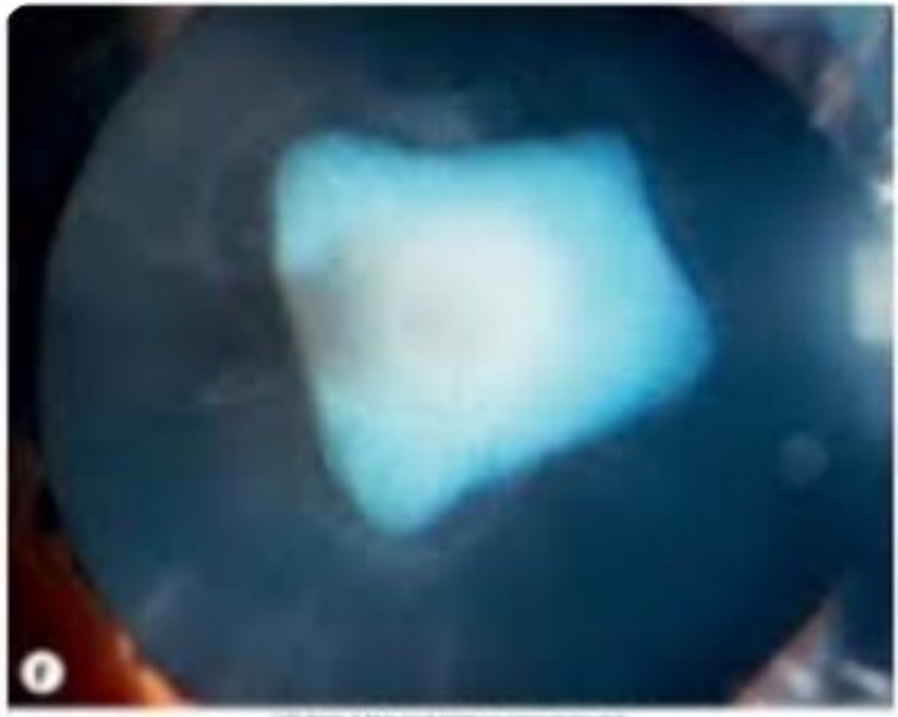
Brushfield Spots

- (MC) metabolic cause for cataract  $\rightarrow$ 
  - Galactosemia
  - Oil drop cataract
  - $\therefore$  urin  $\Rightarrow$  Reducing Substance
- (MCC) reversible cataract



Oil drop cataract

- Anterior lenticonus → Alport's syndrome  
urine ⇒ Blood
- Posterior Lenticonus → Lowe syndrome  
urine ⇒ A.A
- Atopic Dermatitis → Shield cataract



Shield

# Shield cornea → VKC

- Radiation → Glass worker's cataract
- Uveitis → Polychromatic lustre  
    {complicated cataract}
- Drug Causing Cataract ::

- (A) → Au : Gold  
    Amiodarone
- (B) → Busulfan
- (C) → Chloroquine  
    Chlorpromazine  
    [PHENOTHIAZINE]
- (D) → DEXAMETHASONE  
    ↳ Steroid {systemic}
- (E) → Ecothiophate → Miotic  
    /Pilocarpine.

- Anterior polar cataract → Penetrating trauma
- Posterior polar cataract → PHPV {Mittendorf dot}
- Posterior Subcapsular cataract → PSC seen in BSC

↳ Busulfan  
↳ Blower's  
↳ Steroids  
Complicated Cataract  
Cupeliform  
Chloroquine

- Anterior Subcapsular cataract → A: Au  
    Amiodarone  
    Atopic dermatitis  
    C: Chlorpromazine.

## COMPLICATED CATARACT { Rainbow cataract }

- Disturbance in nutrition of lens  
d/t inflammatory / degenerative  
disease of other part of the  
eye like iridocyclitis and  
ciliary body tumours
- C/F: • Breadcrumb appearance  
    • Polychromatic lustre  
    • Early stages → vision is  
        impaired

# Management of Cataract:

No Medical Rx

Surgical Rx:

1) Phacoemulsification → most advanced

↳ ① Incision at Limbus → superior / temporal limbus

② Viscoelastic to maintain the pressure of eye

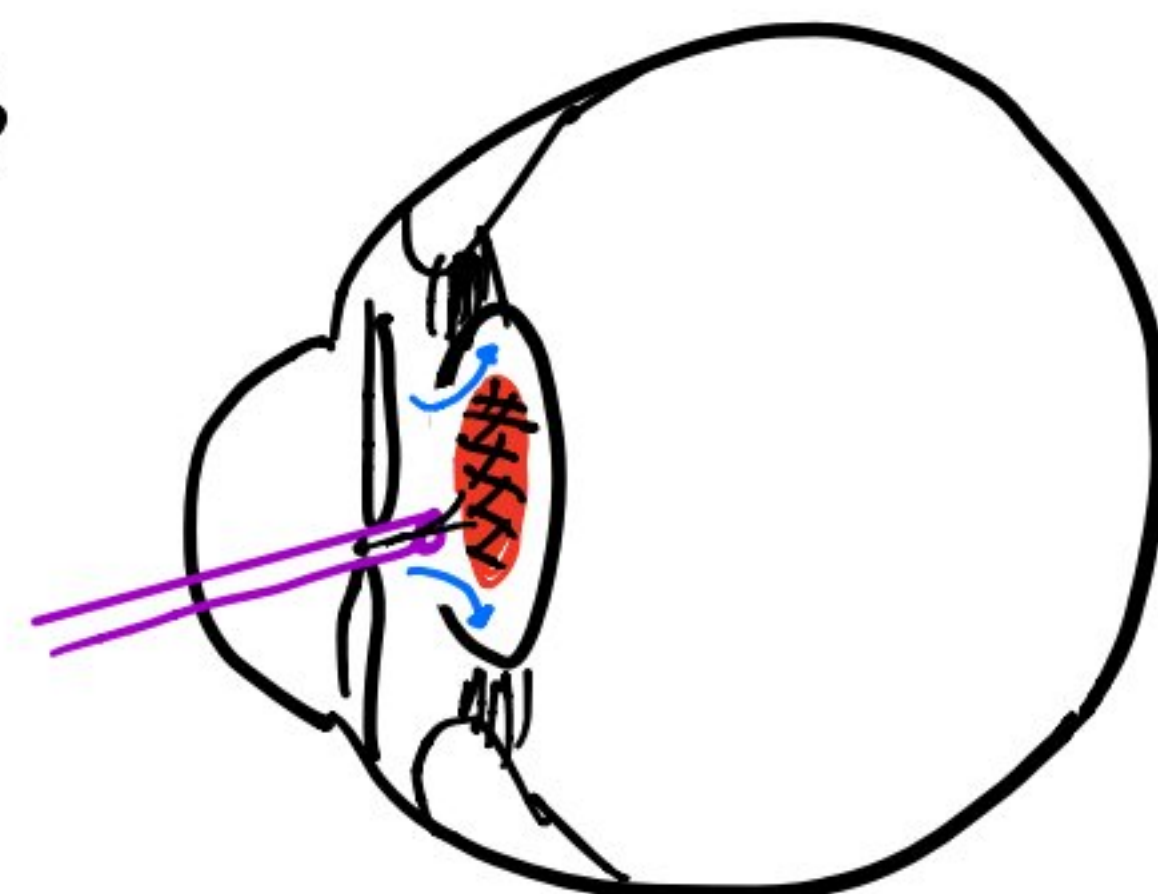
③ Anterior capsulorhexis

↳ Trypan Blue dye

④ Hydrodissection / Hydrodelineation

⑤ Phacoemulsification

⑥ Posterior chamber intraocular lens



2) SICS {small incision cataract surgery}

↳ ① sclerocorneal tunnel  
No sutures required

3) ECCE {Extra capsular cataract extraction}

① ② ③ ④ Same step

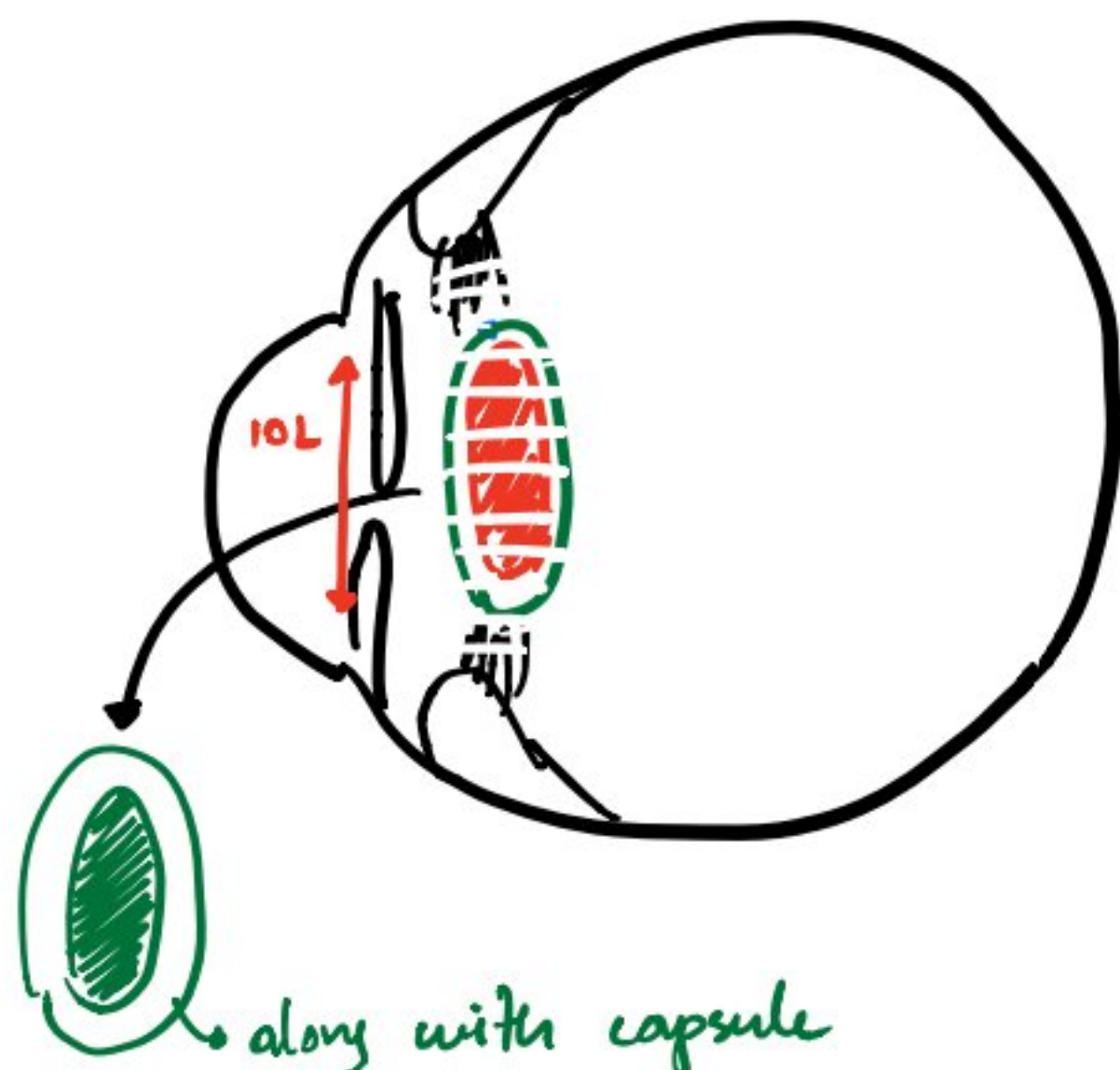
⑤ Removal of nucleus in total → out from limbus → ECCE {suture are → max. req ⇒ 10-0 nylon Astigmatism}

→ out from sclera → SICS

↳ self gating incision, may not require suture

⑥ posterior chamber intraocular lens.

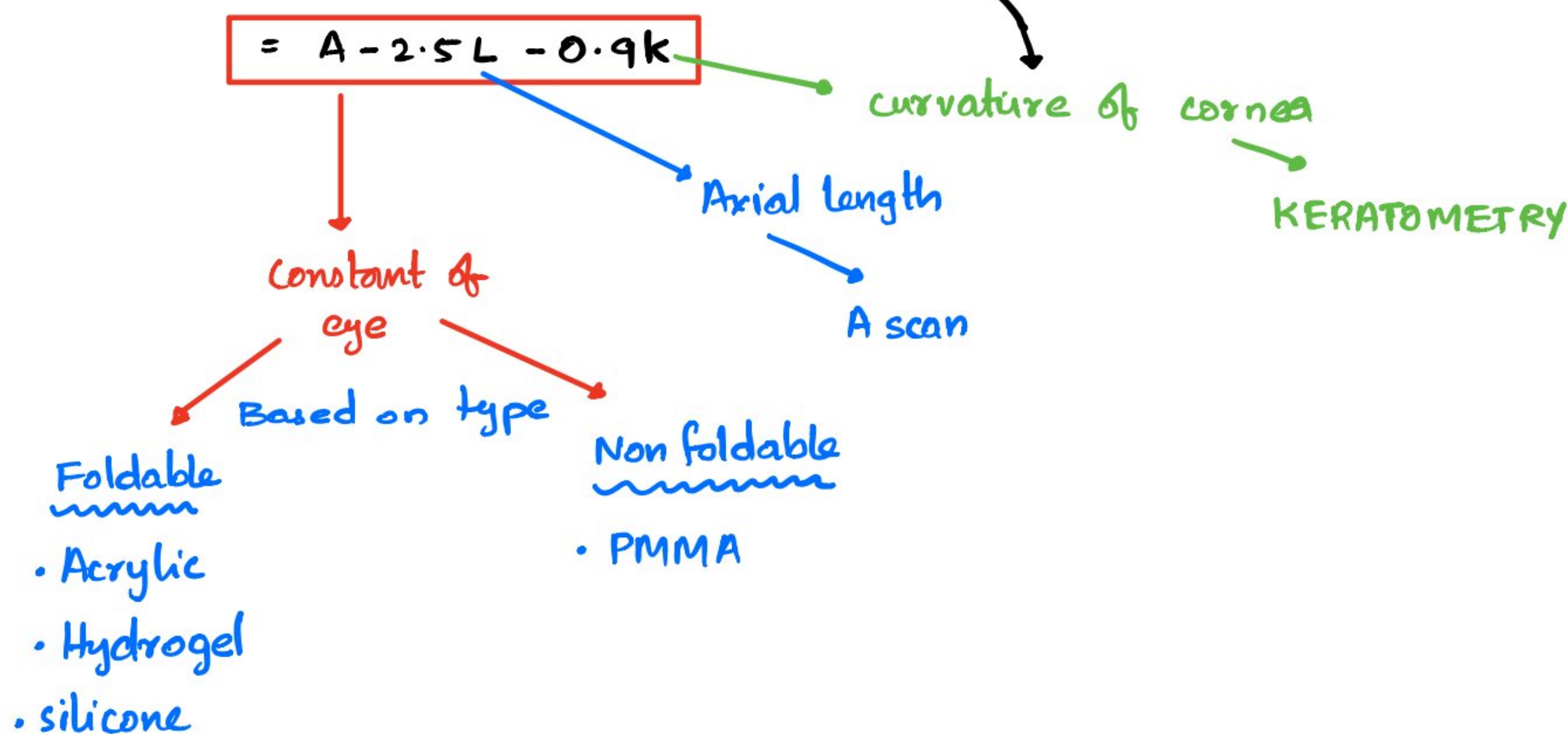
4) ICCE {Intra capsular cataract extraction}



|                  | Phacoret            | Phacoemulsification | SICS            | ECCE            |
|------------------|---------------------|---------------------|-----------------|-----------------|
| Size of Incision | < 3mm               | 3mm                 | 6mm             | 10mm            |
| Astigmatism      | Nil                 | Nil                 | Nil             | Yes             |
| Type of IO L     | Rollable<br>PC-IO L | Foldable<br>PCIO L  | Rigid<br>PCIO L | Rigid<br>PCIO L |
| Sutures          | Nil                 | Nil                 | Nil             | 10-0<br>suture  |

Best Irrigating fluid for cataract surgery is  
BALANCED SALT SOLUT<sup>n</sup> + GLUTATHIONE

IOI power → method: Biometry  
calculation → Formula: SRK



SxOC for congenital cataract ⇒ LENS Aspiration.

(M-C) complication in cataract surgery in children >>> adult

Posterior cataract opacification

↳ To prevent this do::

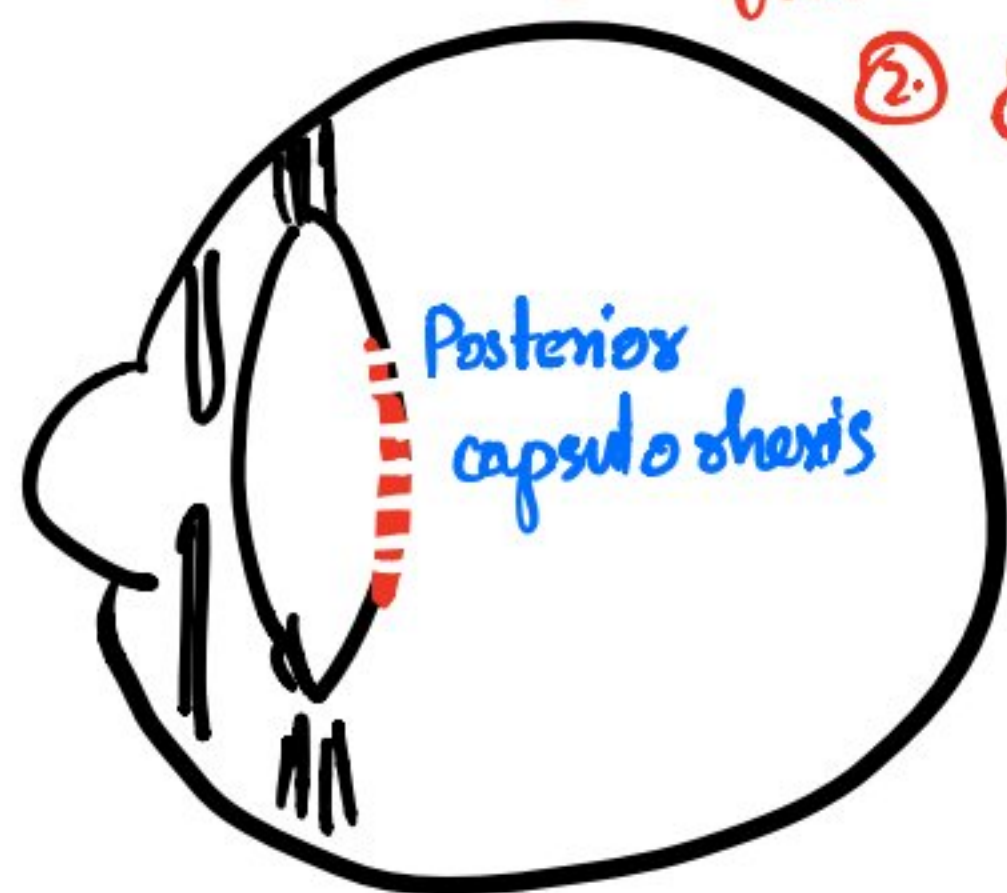
- ⊕ Posterior capsulorhexis
- ⊕ Ant. vitrectomy
- ⊕ IOL implantation

Sx is done ASAP

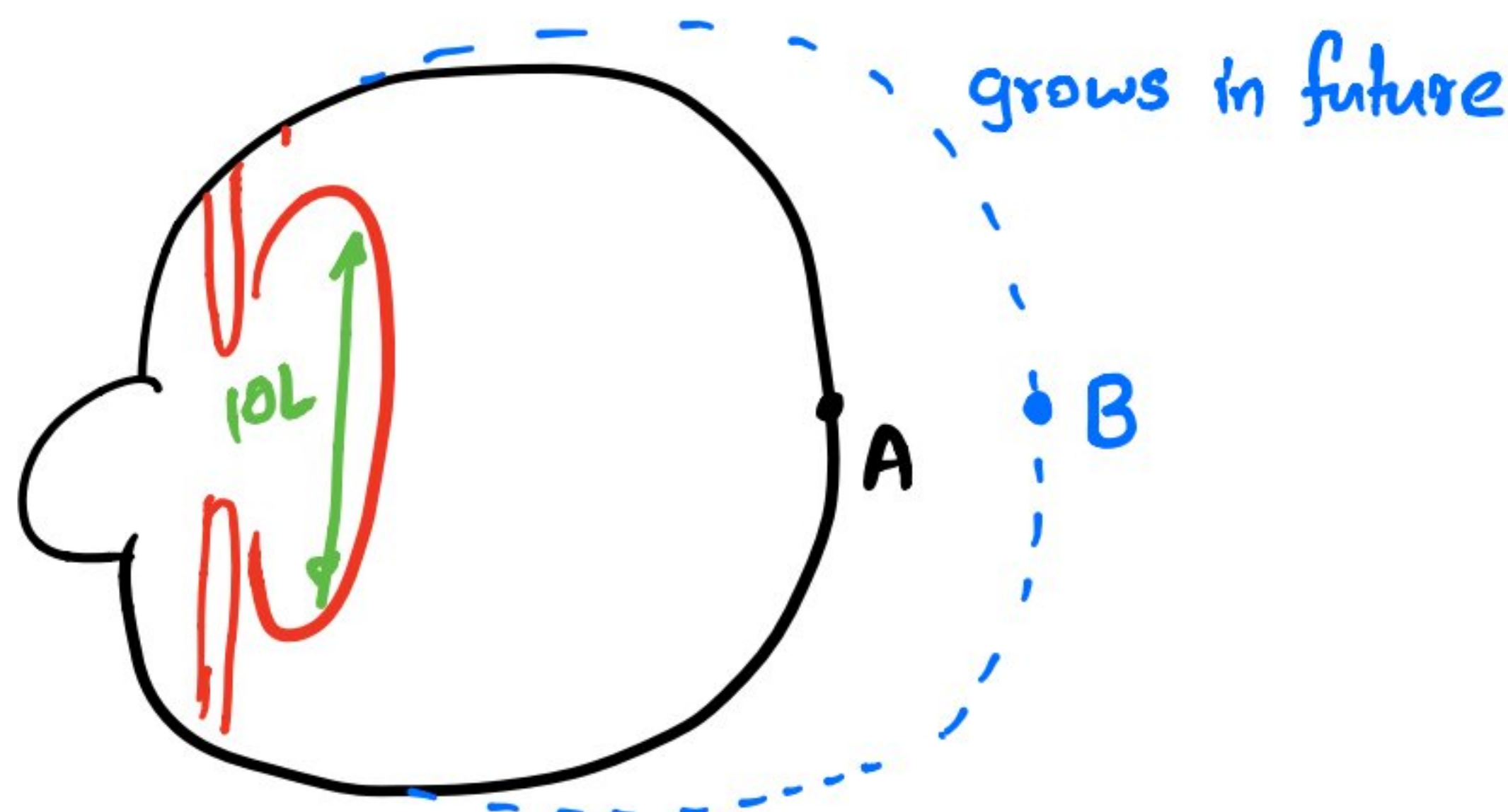
causes of PCO::

① Inflammation

② E cells migrate to post. capsule helped by ant. vitreous



Q: ICCE is C/I in congenital cataract d/t strong Weigert Ligament



Power of IOL in congenital cataract → undercorrection is done so that in future rays converge at Point B.

< 2yr = 20%.

2-8% = 10%.

Q: which extra ocular muscle last affected in anaesthesia → Superior oblique

## COMPLICATION

### Intra Op

- Injury to cornea
- Injury to Iris

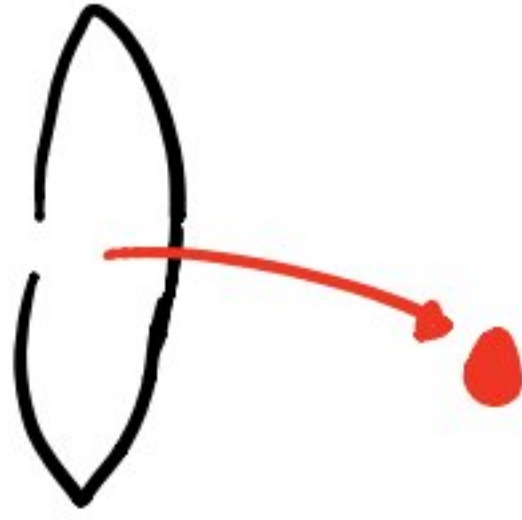
**HYPHAEME**



- Injury to posterior capsule

- nuclear drop

↓  
S/i Posterior  
polar cataract



- Suprachoroidal H'mage

↳ most dangerous complication of cataract surgery

R.F: **Glaucoma**

- old age
- Myopia
- Choroidal sclerosis

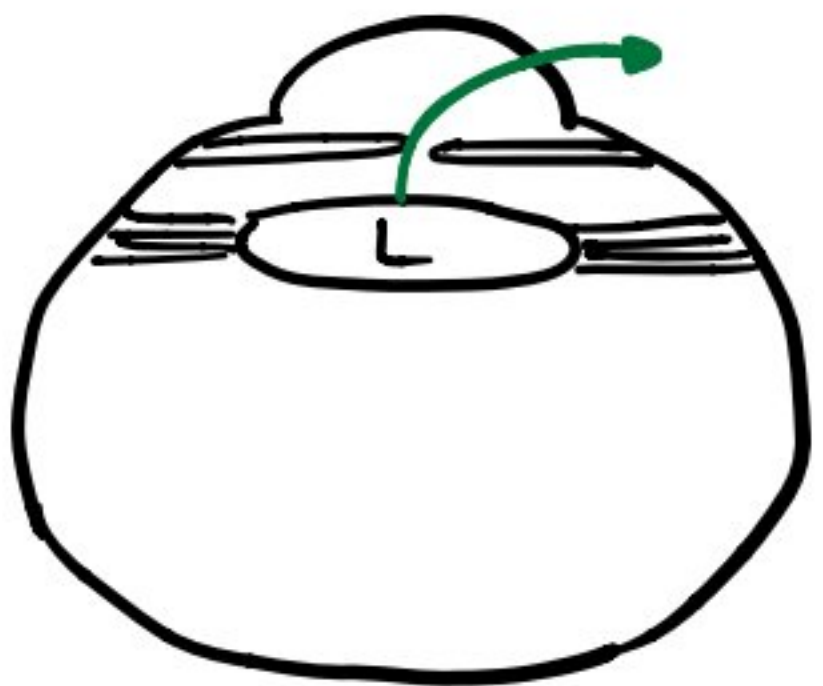
→ sudden rise in IOP

↓  
**Artery rupture**

- Short > Long
- Post ciliary artery rupture

### Steps of Mx:

- 1) Close the incision
- 2) Later, sclerectomy to drain blood



### Prevention of complication

- 5% Povidine - Iodine

Rx: for infection

Topical Antibiotics  
{cannot penetrate vitreous}  
+  
Intravitreal injection of Antibiotics.

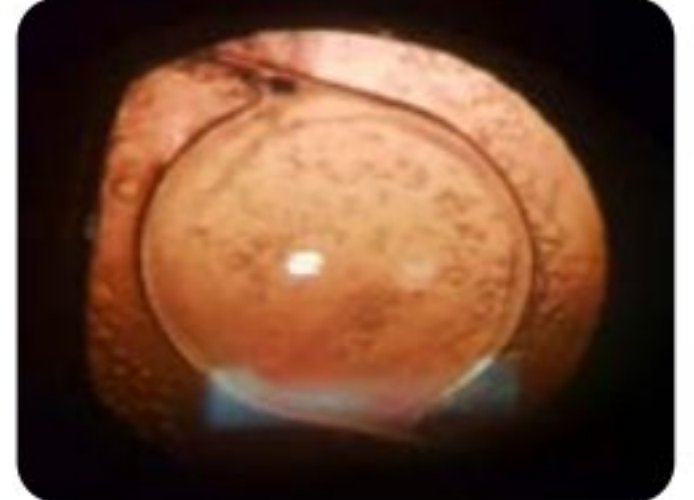
No role in I.V / oral Antibiotics

### Past op

#### Early

- < 6wk
- ↓ IOP = i) wound leak  
ii) Iris prolapse
- ↑ IOP → malignant glaucoma  
pupil block glaucoma

• > 6wk



Elschnig pearl

painless loss of vision + pseudophakic pt. with glare at night while driving

Rx: **Laser Capsulotomy**

↳ Photodisruptive type

ND YAG 1064nm single freq.

### UGH Syndrome:

|          |       |
|----------|-------|
| Uveitis  | ACIOL |
| Glaucoma | ↓     |
| Hyphema  | ICCE  |

### 3) Pseudophakic Bullous Keratopathy

After any surgery there is corneal endothelium cell loss.

### 4) Cystoid Macular Edema

↳ **IRWIN GASS Syndrome**

- ↑ inflammation → break of blood retinal barrier

↓  
CME

### 5) Infection of inner layer of eyeball along with vitreous

↳ ENDOPHTHALMITIS

- occurs after 2-5 days after cataract surgery.
- (MC) organism ⇒ Staph. Epidermidis from skin eyelids
- 96 pt. comes < 1 day ⇒ Staph. aureus
- 96 pt. comes after 1 month  
↳ Propionibacterium acne & c/c3

Surgical Mx : i) < hand movement  
: vision poor  
ii) Take out entire  
vitreous ← Pars plana  
vitrectomy

## ECTOPIA LENTIS:

Subluxation: Partially out of patellar fossa

- M → Marfan's Sx → B/L supro temporal
- A → Aniridia
- T → Blunt Trauma
- H → Homocystinuria → B/L inferonasal  
Hyperlysinemia
- S → Strickler syndrome  
Sulphite oxidase deficiency
- F → Familial cause

Q: (M.C) ocular sign of Marfan's  
↳ Subluxation

Q: Most prominent sign of "  
↳ Megalocornea

## MARFAN'S SYNDROME

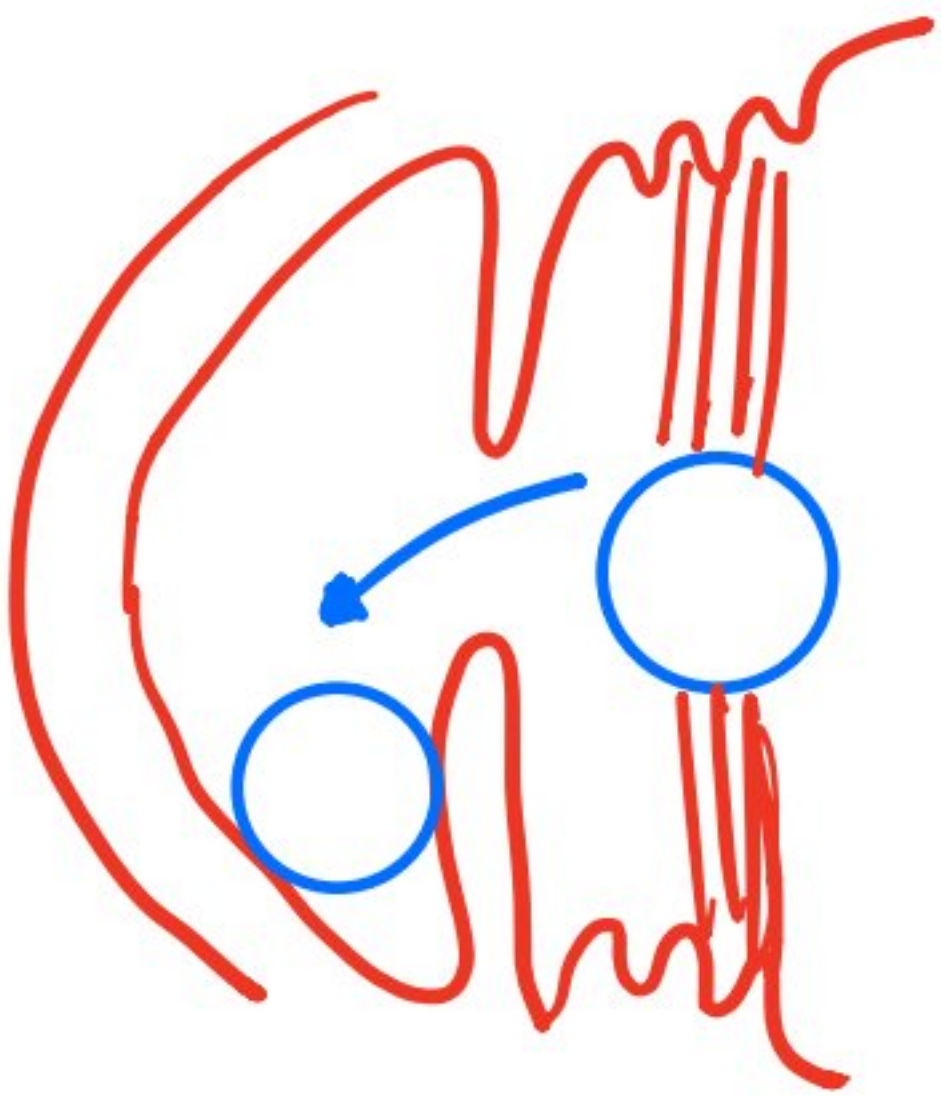
- Eye: B/L suprotemporal subluxation of lens
- Tall, thin
- Long spider like finger

Dislocation: Ant. dislocation = spherical lens  
completely out of the patellar fossa

causes:

1) Blunt trauma → Posterior dislocation

2) Microspherophakia → **WEIL - MARCHESANI SYNDROME**



- Anterior Inferior dislocation
- Inver glaucoma
- Body Habitus: short & stout  
Short & stout  
Short & stubby fingers & toes.



Microspherophakia



Short and stubby hands

## Homocystinuria

- B/L inferonasal subluxation
- Absence of  $\beta$ -cystathionine synthase → {↑ Hyper homocysteinemia} ⇒ Thromboembolic  
Osteoporosis
- Mentally subnormal